Bachelor's thesis

Hidden Markov Models for High Frequency FX Pairs Trading

Regime-gated co-integration strategies on currency tick data

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14th May 2026

Abstract

This thesis asks whether Hidden Markov Models (HMMs) add value to a classical pairs-trading rule on high-frequency currency tick data (Gatev et al. 2006; Vidyamurthy 2004; Avellaneda and Lee 2010; Krauss 2017). Bid–ask tick data from Dukascopy Bank SA (2026) for three currency-pair constructions (AUD/USD–NZD/USD, EUR/NOK–EUR/SEK, and GBP/USD–EUR/USD) are pre-averaged on number of ticks (Jacod et al. 2017) and studied across three independent monthly episodes (August 2024, December 2024, and April 2025), chosen for active price movement under structurally different macro backdrops. The pre-averaged mid-price series is used to build a rolling cointegration spread (Engle and Granger 1987; Dickey and Fuller 1979).

The textbook z-score rule on this spread is the baseline strategy; a Markov-switching AR(1) on the same spread (Hamilton 1989; Krolzig 1997) is the regime model. The HMM is used as a regime gate on top of the baseline, not as an independent forecaster: an AR-gated variant switches the baseline off in a high-volatility drifting regime, and an MS-AR variant widens the entry threshold continuously with the drifting-regime probability. All four strategies (passive Buy & Hold, Baseline, AR, MS-AR) are evaluated under a per-day fit, half-spread plus commission cost model, and walk-forward score (Roll 1984; López de Prado 2018).

I find that, at realistic frictions, the cointegration baseline loses money on every pair-month: the rolling z-score generates frequent round-trips at small per-trade alpha, and transaction costs erode the signal. The HMM gate cuts losses by roughly 75–85% on every pair-month by sharply reducing market exposure, but never crosses into profitability. The continuous-threshold MS-AR variant fails to improve on the binary AR gate, an honest negative finding on the value of the extra parameter (Sharpe 1994; Sortino and Meer 1991).

Passive Buy & Hold dominates all three active strategies on a Sharpe-ratio basis in seven of nine pair–months, including the spring 2025 tariff-driven episode in which a unit long-spread position on EUR/NOK–EUR/SEK earns +528 bps on the same data on which the cointegration rule loses −19,906 bps.

Acknowledgements

I would like to thank my supervisor, Associate Professor Gudmund Horn Hermansen (Department of Mathematics, University of Oslo), for his guidance and support throughout this project.

1 Introduction

Modern financial markets are mathematical objects. The trading pits and discretionary judgement of the past have been replaced by electronic limit-order books, statistical arbitrage engines, and latency-sensitive execution algorithms (Lewis 2014; Aldridge 2013; Biais et al. 2015); prices are no longer just economic summaries of fundamentals but high-dimensional data streams generated by agents, algorithms, inventories, and information shocks (O'Hara 1995; Hasbrouck 2007; Hautsch 2012). The deeper mathematical issue is not speed but non-stationarity: any exploitable structure is weak, transient, and rapidly arbitraged away (Cont 2001), so the trader's question becomes whether a small statistical regularity can be detected, validated, traded, and risk-managed after costs (López de Prado 2018; Aït-Sahalia and Jacod 2014).

The foreign exchange (FX) market is a natural setting for this problem: globally traded, electronically intermediated, and active nearly twenty-four hours a day. At low frequencies, exchange rates are well approximated by continuous diffusions (Bachelier 1900; Black and Scholes 1973; Merton 1973); at the tick level prices move in small discrete increments, quotes update asynchronously, and the price path becomes a sequence of small jumps and microstructural frictions (Roll 1984; Jacod et al. 2017). Figure 1 illustrates this scale dependence: the daily view is approximately diffusive, the hourly view shows localised mean reversion, and the tick view is dominated by discrete bid–ask bounce. Currency pairs with shared economic or geographic exposure are natural candidates for pairs trading, in which one bets on temporary deviations between two related exchange rates reverting to a common stochastic trend (Gatev et al. 2006; Vidyamurthy 2004; Avellaneda and Lee 2010; Krauss 2017); the three constructions used here (AUD/USD vs NZD/USD, EUR/NOK vs EUR/SEK, and GBP/USD vs EUR/USD) span a range of liquidity and dependence regimes.

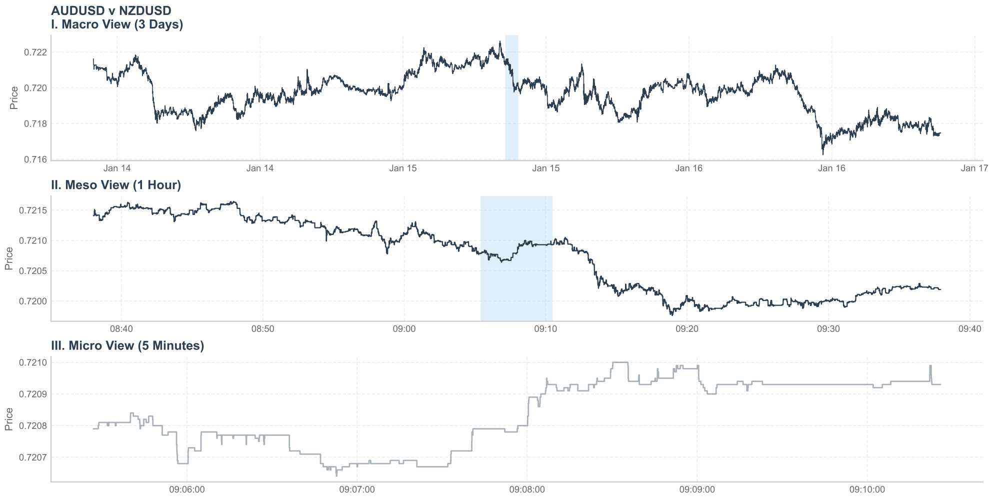

Figure 1: The same FX series at three temporal resolutions. Top: daily closes, approximately diffusive. Middle: hourly bars, localised mean reversion within a slow drift. Bottom: raw tick mid-prices, dominated by discrete bid–ask bounce and asynchronous quote updates.

A second statistical difficulty is non-stationarity within each pair: volatility clusters, correlations rise and fall, and liquidity varies intraday (Engle 1982; Bollerslev 1986; Cont 2001), so a rule that works in one regime may fail in another. This motivates the language of market regimes, persistent latent states each with their own statistical properties, switching by some stochastic mechanism (Hamilton 1989). Because the regime is unobservable and only emitted data are available, the natural modelling tool is a Hidden Markov Model (HMM), in which an observable process is driven by an unobserved Markov chain whose state determines the distribution of the observation (Rabiner 1989; Cappé et al. 2005; Krolzig 1997).

Role of the HMM in this thesis. The exercise is deliberately layered. I first build a single tradable signal, the rolling cointegration spread S_t (Engle and Granger 1987; Dickey and Fuller 1979) standardised into a z-score Z_t, and use it inside a textbook pairs-trading rule (Gatev et al. 2006; Vidyamurthy 2004). This baseline is back-tested under realistic transaction costs and serves as the benchmark against which any added complexity must justify itself. I then fit a Markov-switching AR(1) to the same S_t (Hamilton

1989; Krolzig 1997) to recover a hidden chain K_t representing the active regime, and use the HMM as a regime gate on top of the baseline rather than as an independent forecaster: the trading direction stays dictated by Z_t, and the HMM's role is to suppress or down-weight trading during high-volatility, weak-mean-reversion regimes. In short, the baseline gives the signal and the HMM decides when not to trade. Because the regime-aware strategies inherit the baseline's direction, any difference between them and the baseline is attributable to the gate alone, which is why baseline results are reported before regime-model results. Figure 2 previews this regime recovery on a simulated two-regime AR(1).

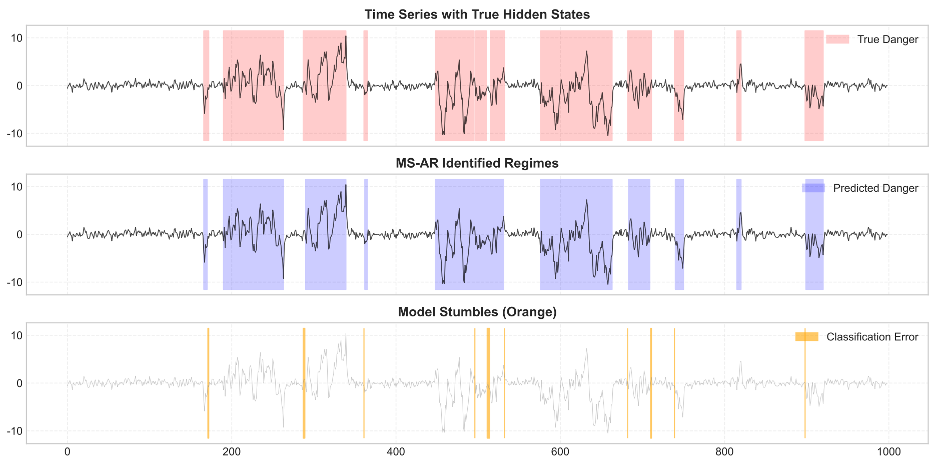

Figure 2: Illustrative HMM regime recovery on a simulated two-regime AR(1). Top: observable series with the true generating regime shaded. Middle: smoothed posterior probability of the high-volatility regime from the Baum–Welch fit (Baum et al. 1970; Rabiner 1989). Bottom: classification error. The hidden chain is recovered from the observable series alone, the same property exploited by the regime gate of Section 4.2 on the real cointegration spread.

Notation and abbreviations. Currency-pair constructions are abbreviated as AUD/NZD (AUD/USD–NZD/USD), EUR/NOK–SEK (EUR/NOK–EUR/SEK), and GBP/EUR (GBP/USD–EUR/USD), with full names on first use and in headers. The four strategies are BH (Buy & Hold), BL (baseline cointegration), AR (binary AR-gated), and MS (continuous MS-AR threshold). Modelling acronyms: HMM (Hidden Markov Model), AR(p) (order-p autoregression), OLS (ordinary least squares), ADF (Augmented Dickey–Fuller), EM (expectation–maximisation), MR (mean-reverting, low-variance regime), DR (drifting, high-variance regime). Reported quantities: SR (Sharpe ratio), MDD (maximum drawdown), UI (Ulcer index), VaR and CVaR (95% Value-at-Risk and Conditional VaR), bps (basis points, 10^{-4}), RT (round-trip trade). A full symbol dictionary closes the appendix in Section 8.8.

Structure. Section 2 describes the data; Section 3 develops the cointegration spread and the baseline rule; Section 4 introduces the AR/HMM machinery and the regime-aware strategies; Section 5 specifies the walk-forward back-test and the performance metrics; Section 6 reports the empirical results; Section 7 concludes.

2 Data and pre-processing

I use high-frequency bid–ask FX tick data from Dukascopy Bank SA (2026), loaded into Python with `pandas` (McKinney 2010) and `NumPy` (Harris et al. 2020). Time-series and Markov-switching estimation use `statsmodels` (Seabold and Perktold 2010); the inner back-test and forward–backward routines are JIT-compiled with `Numba` (Lam et al. 2015) to make the per-day re-fit feasible on tick data. The full data-processing and modelling pipeline, including the back-test driver and the figures reproduced in this thesis, is released as the open-source repository `stk-mat2011` (Furnes 2026). The two legs of a pair are indexed by $i \in \{A, B\}$. The three pair constructions are AUD/USD–NZD/USD, EUR/NOK–EUR/SEK, and GBP/USD–EUR/USD; they span a deliberate range of liquidity, transaction costs, and pair-dependence (Gatev et al. 2006; Krauss 2017).

Rather than running a single continuous back-test, I study three independent one-month sub-samples: August 2024, December 2024, and April 2025. These months were chosen because they cover periods of active price movement on all three pairs while spanning distinct macroeconomic backdrops (summer dollar weakness, year-end positioning, and the spring 2025 tariff-driven episode), which gives three structurally different episodes on which the same machinery can be exercised. Each month is fitted independently per trading day: the rolling cointegration (β_t, α_t), the z-score, and the Markov-switching AR(1) are all re-estimated from scratch every session, so any look-ahead in the smoothed regime posteriors is bounded by a single day.

Bid and ask streams are sorted by timestamp and aligned by a backward as-of merge (Hasbrouck 2007). I write $P_t^{\mathrm{bid},\,i}$ and $P_t^{\mathrm{ask},\,i}$ for the prevailing bid and ask of leg i at time t. The mid-price and basis-point half-spread are

$$\begin{aligned} P_t^{\mathrm{mid},\,i} &= \tfrac{1}{2}\big(P_t^{\mathrm{bid},\,i} + P_t^{\mathrm{ask},\,i}\big), \\ \mathrm{HS}_t^i &= \tfrac{1}{2}\,\frac{P_t^{\mathrm{ask},\,i} - P_t^{\mathrm{bid},\,i}}{P_t^{\mathrm{mid},\,i}} \times 10^4. \end{aligned} \tag{1}$$

HS_t^i approximates the cost of crossing the market for one leg (Roll 1984); since pairs trading enters and exits both legs simultaneously, the combined spread cost is central to the back-test.

Pre-averaging on number of ticks. Tick data are dominated by microstructure noise (bid–ask bounce, discreteness, asynchronous quote updates) that contaminates the observed price relative to the latent efficient price (Roll 1984; Hasbrouck 2007; Hautsch 2012). Following Jacod et al. (2017), I apply a pre-averaging step in tick time: each block of L consecutive ticks is collapsed into a single observation by averaging the mid-prices within the block, so that with $\{P_{t_n}^{\mathrm{mid},\,i}\}_n$ denoting the synchronised tick-time mid-price for leg i, the pre-averaged series at block index k is

$$\widetilde{P}_k^i = \frac{1}{L} \sum_{n=(k-1)L+1}^{kL} P_{t_n}^{\mathrm{mid},\,i}, \qquad k = 1, 2, \ldots \tag{2}$$

The block size L is the same for both legs of a pair, so the pre-averaged legs remain perfectly synchronised. Pre-averaging is preferred here to fixed calendar-time sampling because it produces an observation index that is more nearly homogeneous in microstructure activity and substantially reduces the noise variance of the resulting return series (Jacod et al. 2017; Aït-Sahalia and Jacod 2014; Andersen et al. 2000). All subsequent quantities (log returns, cointegration spread, z-score, and HMM observations) are computed on the pre-averaged series $\widetilde{P}_k^i$. For notational simplicity I continue to write $P_t^{\mathrm{mid},\,i}$ for the price series fed to the models, with the convention that t now indexes pre-averaged blocks rather than individual ticks.

3 Pairs trading with the baseline

This section defines the baseline strategy as a self-contained pairs-trading rule (Gatev et al. 2006; Vidyamurthy 2004); the regime-aware extensions of Section 4 keep its entry direction and only modify when it is allowed to trade.

Cointegration spread. Two non-stationary log-price series $\log P_t^{\mathrm{mid},\,A}$ and $\log P_t^{\mathrm{mid},\,B}$ are cointegrated if some linear combination is stationary (Engle and Granger 1987):

$$S_t \;=\; \log P_t^{\mathrm{mid},\,A} \;-\; \beta\,\log P_t^{\mathrm{mid},\,B}. \tag{3}$$

The hedge ratio β is estimated by the Engle–Granger two-step procedure: OLS regression of $\log P^{\mathrm{mid},\,A}$ on $\log P^{\mathrm{mid},\,B}$, with stationarity of the residuals verified by an Augmented Dickey–Fuller test (Dickey and Fuller 1979; Said and Dickey 1984). High-frequency FX is not globally cointegrated: β drifts with liquidity, session structure, and pair-specific shocks. I therefore estimate β_t by rolling OLS with window W_β. Writing $x_s = \log P_s^{\mathrm{mid},\,B}$, $y_s = \log P_s^{\mathrm{mid},\,A}$, and $\mathcal{W}_t = \{t - W_\beta + 1, \dots, t\}$ with window means $\bar{x}_t, \bar{y}_t$,

$$\beta_t \;=\; \frac{\sum_{s\in\mathcal{W}_t}(x_s-\bar{x}_t)(y_s-\bar{y}_t)}{\sum_{s\in\mathcal{W}_t}(x_s-\bar{x}_t)^2}, \qquad \alpha_t \;=\; \bar{y}_t \;-\; \beta_t\,\bar{x}_t. \tag{4}$$

The cointegration spread and its associated return are

$$\begin{aligned} S_t \;&=\; \log P_t^{\mathrm{mid},\,A} \;-\; \beta_t \log P_t^{\mathrm{mid},\,B} \;-\; \alpha_t, \\ \Delta S_t \;&=\; r_t^A \;-\; \beta_{t-1}\, r_t^B, \end{aligned} \tag{5}$$

with bar log-returns $r_t^i = \log P_t^{\mathrm{mid},\,i} - \log P_{t-1}^{\mathrm{mid},\,i}$. The lag β_{t-1} ensures the hedge ratio is observable when the return is realised.

Rolling z-score and the baseline rule. The baseline reads S_t through a rolling z-score over a window of length W_z,

$$Z_t \;=\; \frac{S_t - \mu_t^S}{\sigma_t^S}, \tag{6}$$

with rolling mean and variance

$$\mu_t^S \;=\; \frac{1}{W_z}\sum_{s=t-W_z+1}^{t} S_s, \qquad \left(\sigma_t^S\right)^2 \;=\; \frac{1}{W_z}\sum_{s=t-W_z+1}^{t}\left(S_s-\mu_t^S\right)^2.$$

With entry threshold z_q and exit threshold z_x, the position $\pi_t \in \{-1, 0, +1\}$ is updated by

$$\pi_t \;=\; \begin{cases} +1 & \pi_{t-1}=0,\; Z_t < -z_q, \\ -1 & \pi_{t-1}=0,\; Z_t > +z_q, \\ \;\;0 & \pi_{t-1}\neq 0,\; |Z_t| \le z_x, \\ \pi_{t-1} & \text{otherwise.} \end{cases} \tag{7}$$

A long-spread position is opened when S_t is sufficiently below its rolling mean and closed when it returns toward zero; the short rule is symmetric (Gatev et al. 2006). For every unit of leg A I trade $\hat{\beta}$ units of leg B, so the resulting portfolio is exposed to the relative move between the two legs rather than to FX direction (Vidyamurthy 2004). Figure 3 illustrates the bookkeeping of a single round-trip on a representative pair.

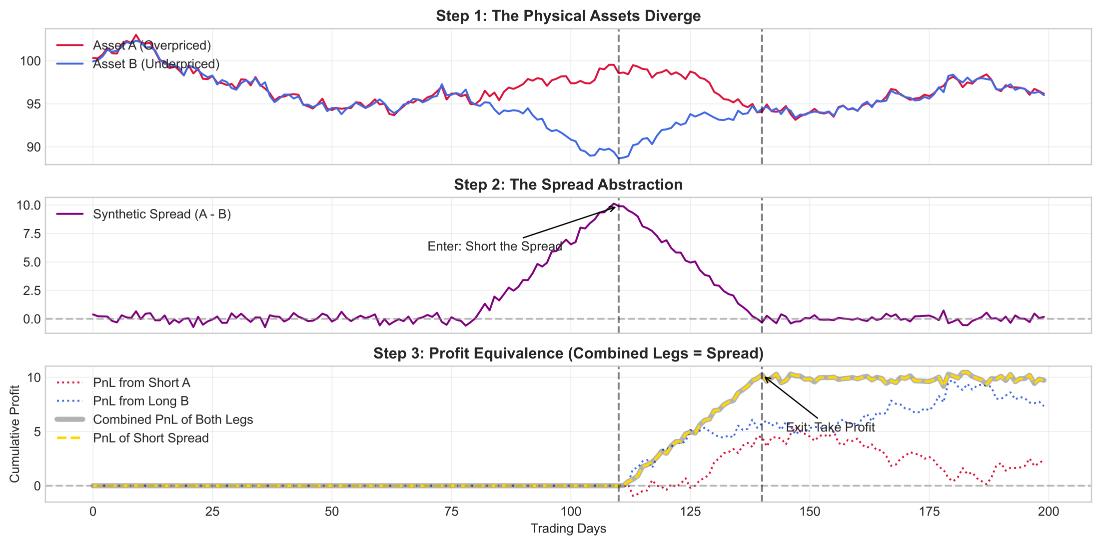

Figure 3: Mechanics of a single pairs-trade. Top: the price path of each leg over the trade window. Bottom: the combined long/short portfolio, which is constructed to be exposed only to the relative move between the two legs and is approximately neutral to the common FX direction. The position is entered when the spread is far from its rolling mean (in z-score units) and closed as the spread reverts.

A do-nothing reference $\pi_t^{\mathrm{BH}} = +1$ for every t (Buy & Hold) earns ΔS_t at each step, pays no transaction costs, and provides a passive drift benchmark. The baseline assumes S_t is governed by a single stationary process, so that fixed (z_q, z_x) correctly characterise "far from equilibrium" for every t; two empirical features of FX tick data violate this, namely positive autocorrelation in squared spread returns (conditional heteroskedasticity (Engle 1982; Bollerslev 1986; Cont 2001)) and directional drift over hundreds of bars during news-driven episodes (Hasbrouck 2007), neither of which the fixed-threshold rule can distinguish. This motivates the regime-aware extensions of Section 4.

4 Regime-aware models

I introduce the AR/HMM machinery on a generic series and then specialise to the spread S_t. The autoregressive model of order p projects a series Y_t on its p most recent past values (Hamilton 1994; Tsay 2010),

$$Y_t \;=\; c \;+\; \phi_1 Y_{t-1} \;+\; \cdots \;+\; \phi_p Y_{t-p} \;+\; \varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \tag{8}$$

with the simplest non-trivial case AR(1),

$$Y_t \;=\; c + \phi Y_{t-1} + \varepsilon_t, \tag{9}$$

having unconditional mean $c/(1-\phi)$ for $|\phi| < 1$. AR(1) captures momentum ($\phi > 0$) or immediate mean reversion ($\phi < 0$) only if (c, ϕ, σ^2) are constant over the sample; to relax that I let the parameters of (9) be piecewise constant, switching with an unobserved discrete-time Markov chain $K_t \in \{1, \ldots, K\}$ (Hamilton 1989; Rabiner 1989; Krolzig 1997; Cappé et al. 2005). The dynamics of K_t are governed by a $K \times K$ transition matrix $\mathbf{P} = (p_{ij})$ and an initial distribution $\nu = (\nu_1, \ldots, \nu_K)^\top$,

$$p_{ij} \;=\; \mathbb{P}(K_t = j \mid K_{t-1} = i), \qquad \nu_k \;=\; \mathbb{P}(K_1 = k), \tag{10}$$

with the row-stochastic constraint $\sum_{j=1}^{K} p_{ij} = 1$ for every i. The Markov assumption is deliberately simple: the future depends on the past only through the current state, yet the framework is rich enough to capture persistence, sudden switching, and regime-dependent parameters within a single probabilistic model.

4.1 Markov-switching AR(1) on the spread

I specialise to S_t from (5) and take $K = 2$. The joint generative model is

$$\begin{aligned} S_t \mid S_{t-1},\, K_t = k \;&\sim\; \mathcal{N}\big(c^{(k)} + \rho^{(k)} S_{t-1},\; (\sigma^{(k)})^2\big), \\ \mathbf{P} \;&=\; \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}, \end{aligned} \tag{11}$$

an AR(1) on the spread with all three parameters $(c^{(k)}, \rho^{(k)}, \sigma^{(k)})$ switching (Hamilton 1989; Krolzig 1997). For $|\rho^{(k)}| < 1$ the regime mean is $\mu^{(k)} = c^{(k)}/(1 - \rho^{(k)})$, yielding the mean-deviation form

$$S_t \;=\; \mu^{(k)} \;+\; \rho^{(k)}(S_{t-1} - \mu^{(k)}) \;+\; \varepsilon_t^{(k)}, \qquad \varepsilon_t^{(k)} \;\sim\; \mathcal{N}(0,\, (\sigma^{(k)})^2). \tag{12}$$

Values of $\rho^{(k)}$ close to zero correspond to fast mean reversion; values close to unity, to near-random-walk behaviour. Before estimation, S_t is winsorised at $\mathrm{median}(S) \pm 4 \cdot \mathrm{std}(S)$ and rescaled by 10^4 for numerical stability.

Estimation, regime choice, and labelling. The HMM parameter set

$$\Theta \;=\; \{\, \nu,\; \mathbf{P},\; c^{(k)},\; \rho^{(k)},\; \sigma^{(k)} : k = 1, \ldots, K \,\} \tag{13}$$

is estimated by maximum likelihood with the EM algorithm (Dempster et al. 1977), which for HMMs is the Baum–Welch procedure (Baum et al. 1970; Rabiner 1989). The E-step computes the smoothed regime posteriors

$$\gamma_t(k) \;=\; \mathbb{P}(K_t = k \mid S_1, \ldots, S_T; \Theta) \tag{14}$$

via the forward–backward algorithm; the M-step updates $(c^{(k)}, \rho^{(k)}, \sigma^{(k)})$ and $\mathbf{P}$ to maximise the expected complete-data log-likelihood, restarted from several random seeds to mitigate convergence to local maxima (Cappé et al. 2005). I fix $K = 2$ throughout, motivated by the regime-switching literature in which a parsimonious two-state model (a low-volatility "normal" regime and a high-volatility "stress" regime) already captures the dominant non-stationarity in returns (Engel and Hamilton 1990; Hardy 2001; J. Bulla and I. Bulla 2006; Hamilton 1989; Krolzig 1997; Cappé et al. 2005; Elliott et al. 1995); in preliminary exploration $K \in \{3, 4\}$ either failed to separate cleanly across seeds or split the high-volatility state into sub-regimes whose entry implications were indistinguishable at the relevant cost level, and the extra parameters would be hard to justify under a BIC-style penalty (Schwarz 1978) at the per-day sample sizes used here. With $K = 2$ I label regimes by innovation variance,

$$k_{\mathrm{MR}} \;=\; \arg\min_k \sigma^{(k)}, \qquad k_{\mathrm{DR}} \;=\; \arg\max_k \sigma^{(k)}, \tag{15}$$

giving a mean-reverting (MR) and a drifting (DR) regime, with $\gamma_t^{\mathrm{MR}} = \mathbb{P}(K_t = k_{\mathrm{MR}} \mid S_1, \ldots, S_T)$ and $\gamma_t^{\mathrm{DR}} = 1 - \gamma_t^{\mathrm{MR}}$. In live trading only past observations are available, so the filtered posterior $\mathbb{P}(K_t = k_{\mathrm{MR}} \mid S_1, \ldots, S_t)$ is used inside the walk-forward back-test while the smoothed posterior is reserved for diagnostic figures.

4.2 Regime-aware strategies

I build two regime-aware strategies on top of the baseline (7); both inherit its entry/exit directions and only change when the rule is allowed to trade.

AR-gated. A binary gate with danger threshold $\delta \in (0, 1)$,

$$G_t \;=\; \mathbb{1}\{\, \gamma_t^{\mathrm{MR}} \;\geq\; 1 - \delta \,\}, \tag{16}$$

modifies the rule to

$$\pi_t \;=\; \begin{cases} 0 & \text{if } G_t = 0 \quad \text{(liquidate, stay flat)}, \\ \pi_t^{\text{baseline}} & \text{if } G_t = 1. \end{cases} \tag{17}$$

The gate is binary: trade exactly like the baseline, or sit in cash. It also applies to existing positions: if I am long the spread and the HMM flags danger, the position goes flat at the next bar.

MS-AR. A continuous, regime-weighted entry threshold,

$$z_q^{\mathrm{eff}}(t) \;=\; \gamma_t^{\mathrm{MR}}\, z_q \;+\; \gamma_t^{\mathrm{DR}}\, z_v, \qquad \text{with } z_v \geq z_q, \tag{18}$$

replaces z_q in (7) by $z_q^{\mathrm{eff}}(t)$. When MR is highly probable, $z_q^{\mathrm{eff}} \approx z_q$ and the strategy trades like the baseline; when DR dominates, a larger $|Z_t|$ deviation is required before entering.

Table 1 summarises the four strategies compared throughout the rest of the thesis. All three active rules share the cointegration spread S_t and its z-score Z_t and differ only in whether and how γ_t^{MR} enters the decision.

Table 1: The four strategies compared in this thesis. The “Gate” column gives the indicator multiplying the baseline position π_t^{baseline} before it is taken; a value of 1 means the baseline is always allowed to trade.

Strategy	Always long?	Entry threshold	Gate	Reads HMM?
Buy & Hold	Yes	n/a	n/a	No
Baseline	No	z_q	1	No
AR	No	z_q	$\mathbb{1}\{\gamma_t^{\mathrm{MR}} \geq 1 - \delta\}$	Yes (binary)
MS-AR	No	$\gamma_t^{\mathrm{MR}} z_q + \gamma_t^{\mathrm{DR}} z_v$	1	Yes (continuous)

5 Back-test protocol and metrics

Costs and walk-forward. Every position change in leg i pays the half-spread HS_t^i of (1) plus a fixed commission (Aldridge 2013; López de Prado 2018), so the round-trip cost of one pair-position is approximately $\mathrm{HS}^A + \hat{\beta}\,\mathrm{HS}^B$ in basis points. Strategies are evaluated walk-forward: parameters Θ and hyperparameters $(W_\beta, W_z, z_q, z_x, z_v, \delta)$ are fitted on a training segment and applied to a disjoint out-of-sample test segment, with windows slid forward (López de Prado 2018; Hautsch 2012).

Performance metrics. Let R_t denote the strategy return at bar t, $\mu_R = \mathbb{E}[R]$, $\sigma_R = \sqrt{\mathrm{Var}(R)}$, V_t the portfolio value with running peak $V_t^\star = \max_{s\le t} V_s$, and R_f a benchmark risk-free rate (set to zero). Statistical metrics describe the shape of the return distribution; financial metrics translate it into capital outcomes (Tsay 2010; Cont 2001). The annualised volatility, skewness, and excess kurtosis are

$$\sigma_{\mathrm{ann}} = \sigma_R\sqrt{T}, \qquad \mathcal{S} = \frac{\mathbb{E}[(R-\mu_R)^3]}{\sigma_R^3}, \qquad \mathcal{K} = \frac{\mathbb{E}[(R-\mu_R)^4]}{\sigma_R^4}. \tag{19}$$

With $\alpha = 0.95$, the Value-at-Risk, Conditional VaR, and tail-fatness ratio are (Artzner et al. 1999)

$$\mathrm{VaR}_\alpha = -Q_{1-\alpha}(R), \qquad \mathrm{CVaR}_\alpha = -\,\mathbb{E}\big[R \mid R \le -\mathrm{VaR}_\alpha\big], \qquad \mathrm{TF} = \frac{\mathrm{CVaR}_\alpha}{\mathrm{VaR}_\alpha}. \tag{20}$$

The drawdown family uses $D_t = (V_t^\star - V_t)/V_t^\star$, with maximum drawdown and Ulcer index

$$\mathrm{MDD} = \max_t D_t, \qquad \mathrm{UI} = \sqrt{\frac{1}{N}\sum_{t=1}^{N} D_t^2}\,, \tag{21}$$

where UI is the Ulcer index of Martin and McCann (1989) and the maximum drawdown is treated in Magdon-Ismail and Atiya (2004). Risk-adjusted ratios relate excess return to risk: the Sharpe ratio (Sharpe 1994) divides by total volatility, the Sortino ratio (Sortino and Meer 1991) divides by downside deviation σ_d, and the Calmar ratio divides annualised return AR by the worst observed drawdown,

$$\mathrm{SR} = \frac{\mu_R - R_f}{\sigma_R}, \qquad \mathrm{Sortino} = \frac{\mu_R - R_f}{\sigma_d}, \qquad \mathrm{Calmar} = \frac{\mathrm{AR}}{\mathrm{MDD}}. \tag{22}$$

Trade-level diagnostics (win rate, market exposure, profit factor, payoff ratio) summarise frequency and size; see Section 6 for their interpretation.

6 Empirical results and discussion

All numbers are net of the half-spread-plus-commission cost model of Section 5, with the per-leg half-spread of (1) paired with a fixed commission of 0.05 bps per flip and a per-day fit. For each of the three pair constructions I report three monthly episodes side by side (August 2024, December 2024, and April 2025) to keep cross-episode robustness visible at all times (López de Prado 2018). The full pair–month tearsheets, daily PnL ledgers, and per-pair figure-level diagnostics are in Appendix 8; the discussion below cites the appendix tables directly rather than reproducing them.

6.1 Headline numbers

Table 2 reports total PnL in basis points for the four strategies across the nine pair–month cells; Table 3 reports trading activity and the risk-adjusted Sharpe ratio for the three active rules on the same cells. The full back-test grid (return, risk, risk-adjusted, trading, and distributional metrics) is in Tables 4–6 of Appendix 8.1, and the raw day-by-day ledgers are in Appendix 8.7 (Tables 7–9).

Table 2: Total PnL by pair, month, and strategy, in basis points, net of half-spread plus 0.05 bps commission. BH is Buy & Hold, BL the baseline cointegration rule, AR the binary AR-gated variant, and MS the continuous MS-AR threshold.

Pair	**Month**	**BH**	**BL**	**AR**	**MS**
AUD/USD–NZD/USD	Aug. 2024	−45.8	−966.5	−179.3	−206.1
	Dec. 2024	−39.3	−776.5	−136.7	−158.0
	Apr. 2025	+155.2	−2521.4	−652.2	−731.3
EUR/NOK–EUR/SEK	Aug. 2024	−148.5	−6277.4	−1816.7	−2490.5
	Dec. 2024	+60.6	−4473.8	−1592.8	−2070.8
	Apr. 2025	+527.7	−19906.4	−8103.5	−9487.3
GBP/USD–EUR/USD	Aug. 2024	+65.1	−396.1	−48.6	−60.1
	Dec. 2024	+25.1	−730.4	−69.6	−75.8
	Apr. 2025	+15.9	−1509.2	−350.6	−405.9

Table 3: Trading activity and Sharpe ratio for the three active strategies, by pair and month. "RT" is the number of round-trip trades; "Exp. %" is the percentage of bars in which the strategy holds a non-zero position; Sharpe is annualised (Sharpe 1994; Lo 2002).

		Round-trips			**Exposure (%)**			**Sharpe**		
Pair	**Month**	BL	AR	MS	BL	AR	MS	BL	AR	MS
AUD/USD–NZD/USD	Aug. 2024	193	39	45	31	6	8	−33.8	−18.8	−14.3
	Dec. 2024	186	38	41	27	5	6	−36.9	−21.4	−18.2
	Apr. 2025	434	140	148	35	11	12	−58.6	−36.5	−36.4
EUR/NOK–EUR/SEK	Aug. 2024	606	137	158	35	7	9	−64.5	−45.7	−37.5
	Dec. 2024	435	158	181	35	12	15	−69.2	−58.4	−48.9
	Apr. 2025	1116	495	545	36	15	17	−126.1	−104.9	−104.5
GBP/USD–EUR/USD	Aug. 2024	281	79	83	34	7	9	−17.2	−5.9	−5.8
	Dec. 2024	325	58	63	35	6	7	−32.6	−12.3	−9.2
	Apr. 2025	590	154	161	36	9	10	−45.7	−32.6	−29.2

Three patterns are visible in Tables 2–3 and recur in every diagnostic below. The Baseline loses money on every pair–month, generating between 186 and 1,116 round-trips and posting Sharpe ratios from −17 on the cheapest cell to −126 on the highest-spread one: the rule earns a small per-trade alpha but pays the half-spread plus commission on every flip, and at this resolution the alpha does not clear the friction (Krauss 2017; Roll 1984; López de Prado 2018; Caldeira and Moura 2013). The regime gate cuts exposure by roughly a factor of four to six and cumulative loss by roughly 75–85%, but never turns the strategy positive; the AR gate pulls exposure from 27–36% down to 5–15%, and the MS-AR variant is 1–3 percentage points more permissive than AR on every cell. Finally, passive Buy & Hold dominates the active rules on a Sharpe basis in seven of the nine cells, and is the only positive number in any column on EUR/NOK–EUR/SEK in December 2024 and April 2025 and on GBP/USD–EUR/USD in all three months.

6.2 Pair-level diagnostics and heavy tails

The three pair constructions span a deliberate range of liquidity (Roll 1984; Hasbrouck 2007): GBP/USD–EUR/USD is the cheapest, AUD/USD–NZD/USD is intermediate, and EUR/NOK–EUR/SEK is materially more expensive. All six leg-level return series are strongly non-Gaussian, with high kurtosis and significant Ljung–Box statistics on squared returns, consistent with conditional heteroskedasticity (Engle 1982; Bollerslev 1986; Cont 2001); the rolling spread-volatility panels in Appendix 8.3 make the volatility clustering visible directly, and the per-bar period-return panels in Appendix 8.4 document the same fat-tailed behaviour at the trade-relevant horizon. Rolling Augmented Dickey–Fuller tests (Engle and Granger 1987; Dickey and Fuller 1979; Said and Dickey 1984) with Newey–West standard errors (Newey and West 1987), computed via `statsmodels` (Seabold and Perktold 2010), reject the no-cointegration null intermittently for AUD/USD–NZD/USD and GBP/USD–EUR/USD; EUR/NOK–EUR/SEK rejects globally but with a long implied half-life, which makes it economically weak for high-frequency mean reversion (Krauss 2017; Caldeira and Moura 2013).

The shape of the per-bar spread-return distribution makes the friction story concrete. Figure 4 reports, for AUD/USD–NZD/USD across the three episodes, the spread-return density on linear (top row) and log (bottom row) y-axes: the linear panels show a sharply peaked, near-zero-centred bulk, while the log panels reveal Gaussian-incompatible tails at three or more standard deviations (Cont 2001). The corresponding grids for the other two pairs are in Appendix 8.5; on EUR/NOK–EUR/SEK the log-tail panels are visibly the widest, a precursor to the largest active-strategy losses below.

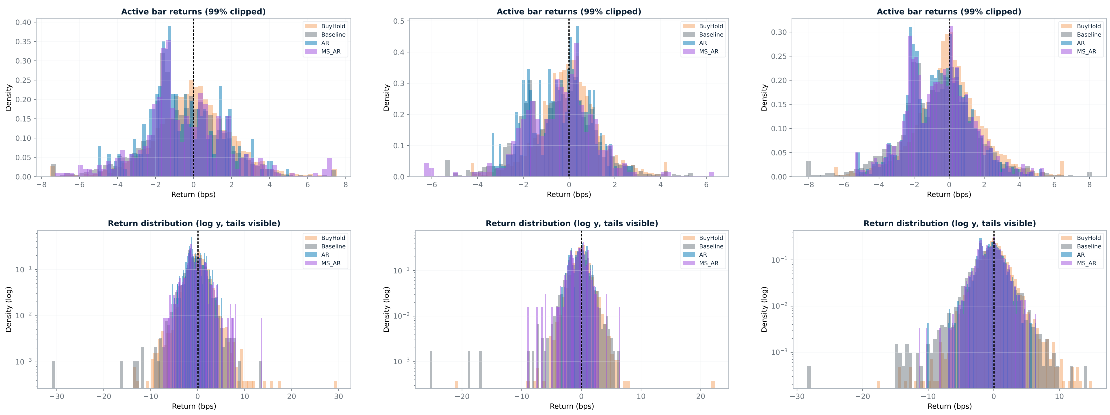

Figure 4: AUD/USD–NZD/USD spread-return distributions across the three episodes. Columns: August 2024 (left), December 2024 (centre), April 2025 (right). Top row: linear y-axis; bottom row: log y-axis exposing the Gaussian-incompatible tails of Cont (2001).

6.3 Baseline and HMM gate side by side

Figure 5 shows the cumulative profit-and-loss of the four strategies stacked across the three months for AUD/USD–NZD/USD; the corresponding panels for the two other pairs are collected on a single symlog grid in Appendix 8.2. Passive Buy & Hold drifts mildly; the Baseline (rolling z-score with $z_q = 2.0$, $z_x = 0$) bleeds steadily; the HMM-aware strategies AR and MS-AR sit between the two, losing less than the Baseline but never crossing into profit.

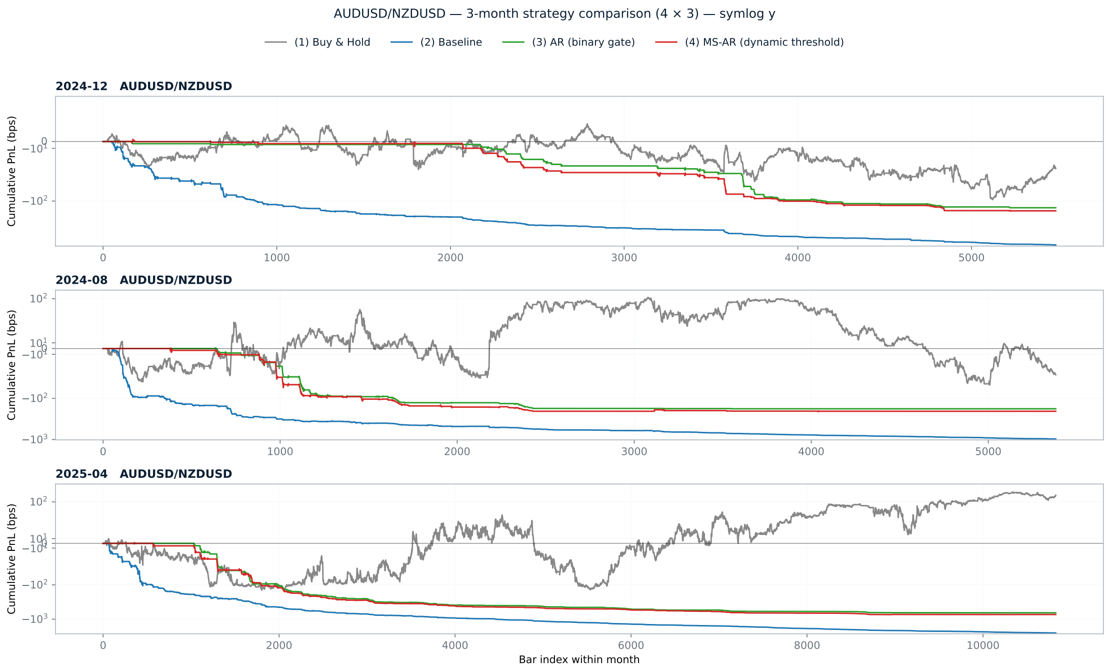

Figure 5: AUD/USD–NZD/USD, three-month strategy comparison on a symmetric-log y-axis. Each panel stacks the four cumulative-PnL curves for one month under the half-spread + 0.05 bps commission cost model. The Baseline (blue) leaks steadily; the AR gate (green) and MS-AR threshold (red) trace a much flatter, less negative trajectory; Buy & Hold (grey) is positive in April 2025.

The Baseline result has three forces behind it. The rolling z-score generates between roughly 200 and 1,100 round-trips per month per pair, and each round-trip pays $\mathrm{HS}_t^A + \hat{\beta}\,\mathrm{HS}_t^B$ plus commission (Roll 1984; Aldridge 2013; López de Prado 2018). The per-trade alpha is small relative to that friction: the average winning trade is 1–1.5 bps on AUD/USD–NZD/USD and GBP/USD–EUR/USD and slightly larger on EUR/NOK–EUR/SEK, with comparable average losses (Appendix 8.1), and the profit factor sits between 0.21 and 0.76 in every cell, the standard cost-of-mean-reversion-on-FX result of Krauss (2017) at tick resolution. And the loss is heavily left-skewed and fat-tailed: Baseline skewness ranges from −0.29 to −13.0 and kurtosis from 12.5 to 308.6, so a handful of trade-level disasters dominate cumulative outcomes; April 2025 EUR/NOK–EUR/SEK is the extreme case, with 1,116 round-trips, −19,906 bps realised, and a profit factor of 0.21.

The HMM gate inherits the Baseline's entry direction and only changes when the rule is allowed to trade. The per-day Baum–Welch fits (Baum et al. 1970; Dempster et al. 1977; Rabiner 1989; Elliott et al. 1995) cleanly separate the two regimes by innovation variance on every pair–day (Figures 14–16 in Appendix 8.6): the MR state has the smaller $\sigma^{(k)}$ and higher persistence p_{ii}, the DR state the larger $\sigma^{(k)}$ and smaller p_{ii}, and both means $\mu^{(k)}$ are close to zero, as required for the cointegration interpretation of S_t. This clean separation translates into a uniform reduction in trading activity: the AR gate fires $G_t = 0$ in the drifting regime and shuts the Baseline off, pulling exposure from 27–36% down to 5–15%. The MS-AR threshold of (18) softens the binary switch into a continuous threshold interpolating between z_q and z_v, which is slightly more permissive, raising exposure by 1–3 percentage points and round-trip count by 5–10. The same hierarchy is visible across risk measures: annualised volatility, max drawdown, and the Ulcer index of Martin and McCann (1989) all fall in the order BL > MS > AR on every pair–month (Appendix 8.1). What the gate does not do is convert reduced exposure into positive risk-adjusted return: the AR Sharpe is strictly less negative than the Baseline Sharpe everywhere, by a factor of two to five, but it never crosses zero. Once one conditions on the rare drifting-regime entries the surviving sample is dominated by tail events and the kurtosis of the gated strategies is in fact larger than the Baseline's. The HMM, on this data, buys insurance, not alpha (Sharpe 1994; Sortino and Meer 1991; Lo 2002).

The continuous-threshold MS-AR adds one free hyperparameter, z_v, on top of the binary AR gate of (16), with the motivation of exploiting in-between values of γ_t^{MR} rather than throwing them away. The additional flexibility does not pay for itself: on every pair–month in Table 2, MS-AR loses slightly more than AR (for example -731.3 vs -652.2 bps on AUD/USD–NZD/USD April 2025), because at $z_v = 3$ the continuous threshold admits a handful of marginal entries that the binary gate would have suppressed, and at this cost level those entries are net-negative; on two of nine cells MS-AR beats AR on Sharpe by a small margin, consistent with the warning of Cappé et al. (2005) that smoothing a regime decision rule is only beneficial when the marginal entries it admits are profitable in expectation. A BIC-style penalty (Schwarz 1978) on z_v, applied via a hyperparameter search such as Optuna (Akiba et al. 2019), would already select against MS-AR here.

6.4 Discussion

The four strategies tell a coherent story. Buy & Hold is the only one with a positive cumulative PnL in a majority of pair–months and the only one not paying a tick-level cost; its positive realisations (the largest of which is +527.7 bps on EUR/NOK–EUR/SEK April 2025, on the same data on which the Baseline loses $-19{,}906.4$ bps) reflect persistent directional drift in the spread rather than any traded signal (Tsay 2010; Gatev et al. 2006; Caldeira and Moura 2013). The Baseline is the textbook cointegration rule and the worst-performing strategy on every pair–month: its per-trade alpha is in the right direction (the win rate is close to 40% and several daily ledgers in Appendix 8 contain individual gross-positive days) but does not clear the round-trip half-spread, reproducing the headline of Krauss (2017) on tick-level FX. AR delivers the gate's promise (a uniform four-to-six-fold cut in exposure and round-trip count, a roughly three-quarters to four-fifths reduction in cumulative loss, proportionally smaller drawdowns and CVaRs); whether one calls it an improvement depends on the loss function, since on Sharpe it is still unambiguously bad. MS-AR substitutes a continuous threshold for the binary switch and is mildly worse on PnL but marginally better on Sharpe in two of nine cells, so the extra parameter z_v does not earn its keep at this cost level.

The qualitative pattern was partly anticipated: that high-frequency FX is cost-dominated is well known (Aldridge 2013; Hautsch 2012; Krauss 2017), and the conjecture in Section 1 that the HMM "decides when not to trade, not what to trade" aligns with the data. Less expected is the size of the Buy & Hold advantage in directional regimes such as April 2025 and the cleanness of the pair-level cost ordering across all nine cells: GBP/EUR (cheapest, smallest loss), AUD/NZD (intermediate), EUR/NOK–SEK (most expensive, largest loss). That ordering is monotone in every month and is the cleanest causal statement I can make from this data, the Baseline's loss being to leading order the round-trip half-spread times the number of round-trips. Posed as in Section 1, the question of whether the regime gate adds value relative to the baseline or mainly reduces exposure has an unambiguous answer: the gate reduces exposure, but the Baseline does not carry enough net alpha at this cost level for that reduction to translate into positive risk-adjusted returns. The HMM is therefore best read as a costed risk-reduction device that locates the boundary at which classical pairs-trading reasoning runs out of room on FX tick data, and the constructive next step is to pair the regime posteriors with an entry signal that is itself profitable after costs (a momentum signal in the spirit of Jegadeesh and Titman (1993) and Avellaneda and Lee (2010), for example) rather than asking the HMM to rescue a baseline that is not (Krauss 2017).

7 Conclusion

This thesis has tested whether Hidden Markov Models add value to a classical pairs-trading rule on high-frequency FX tick data. A single baseline signal, the rolling z-score of the cointegration spread (Engle and Granger 1987; Gatev et al. 2006; Vidyamurthy 2004; Caldeira and Moura 2013), is first treated as a self-contained strategy and back-tested under a realistic half-spread plus 0.05 bps commission cost model with a per-day fit (Aldridge 2013; López de Prado 2018; Seabold and Perktold 2010; Lam et al. 2015); a Markov-switching AR(1) is then layered on as a regime gate (Hamilton 1989; Rabiner 1989; Krolzig 1997; Elliott et al. 1995), with two concrete instantiations (a binary AR-gated rule and a probability-weighted MS-AR threshold) that leave the baseline's directional entry rule unchanged. All four strategies are compared on the same nine pair–month cells (three currency pairs across August 2024, December 2024, and April 2025).

Four findings emerge. First, the cointegration baseline is unprofitable at realistic frictions on every pair–month: the rolling z-score generates a stream of round-trips at small per-trade alpha and bleeds net of cost, with Sharpe ratios from -17 on the cheapest pair–month to -126 on the highest-spread one (Tables 2–3), and the pair-level magnitude of damage is monotone in the per-leg half-spread, the cleanest causal statement I can make from the data (Krauss 2017; Roll 1984; Caldeira and Moura 2013). Second, the HMM gate is a defensive filter and not a return enhancer: both regime-aware strategies cut exposure (typically from around 30% down to 5–17%), round-trips (by a factor of three to five), annualised volatility, and the Ulcer index of Martin and McCann (1989), with cumulative loss roughly 75 to 85% smaller on every pair–month, but the strategy never crosses into profit. Third, the continuous MS-AR threshold of (18) does not improve on the binary AR gate of (16): it is mildly worse on cumulative PnL across the board and marginally better on Sharpe in only two of nine cells, so the extra parameter z_v does not earn its keep at this cost level and a BIC-style penalty (Schwarz 1978; Akiba et al. 2019) would already select against it. Fourth, passive Buy & Hold dominates the three active strategies in seven of the nine cells; the most striking comparison is EUR/NOK–EUR/SEK in April 2025, where passive drift earns +527.7 bps on the same spread the cointegration rule loses $-19{,}906.4$ bps on. The Baum–Welch fits (Baum et al. 1970; Dempster et al. 1977) are interpretable in every case (Appendix 8.6), with well-separated innovation variances and persistences close to one for the mean-reverting state, so the negative result on profitability cannot be blamed on a misbehaving model.

These findings frame the central contribution: not a claim that HMMs universally outperform simpler models (they do not), but a costed, per-day-fit test of when regime awareness helps and when it fails on noisy, costly, non-stationary tick data (Cont 2001; Hautsch 2012). The exercise is restricted to three pair constructions, three one-month sub-samples, $K = 2$ regimes, unit-size positions, a half-spread-plus-commission cost model that omits latency and market impact (Aldridge 2013), and an in-sample smoothed posterior for the gates, so the negative finding is best read as a statement about the textbook z-score rule under realistic FX frictions, not about HMMs as such; the three months were also chosen for active price movement under structurally different macro backdrops, which biases the sample toward periods in which a passive long-spread position can drift directionally, so the Buy & Hold advantage should be read as a property of the chosen episodes rather than a universal claim.

Three extensions follow. The most direct is to pair the regime posteriors with a different entry signal in the drifting regime, for example a momentum signal in the spirit of Jegadeesh and Titman (1993) and Avellaneda and Lee (2010), rather than asking the HMM to rescue a mean-reversion signal that is already cost-dominated; the negative result tells future work where not to spend complexity budget. A second is to pre-screen the pair universe with the baseline back-test, since the cost-ordering of the three pairs is so clean that a single Baseline back-test eliminates the high-spread pair regardless of any regime gate (Krauss 2017; Caldeira and Moura 2013). A third is to replace the per-day in-sample smoothing with a true walk-forward filter using only past observations; since the filter is a strict subset of the smoother's information set, that step can only lower the gate's value further and tightens rather than weakens the negative conclusion (Cappé et al. 2005; Elliott et al. 1995). A further direction is an HMM–GARCH hybrid (Bollerslev 1986; Engle 1982) that lets the within-regime conditional variance evolve smoothly, a natural fit for the heavy-tailed distributions of Figure 4. The full reproducible implementation (data pipeline, back-test driver, daily ledgers, and per-pair tearsheets) is released as the open-source repository `stk-mat2011` (Furnes 2026) as a starting point.

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8 Appendix

This appendix collects the quantitative tearsheets, figure-level diagnostics, and raw daily ledgers that accompany the empirical results of Section 6. Where the body of the thesis presents headline numbers and a small number of representative figures, the appendix is the place to look for the full back-test grid, the per-pair regime dynamics, and the trade-by-trade behaviour that the prose summarises. To keep the page count manageable I report each figure-level diagnostic for the August 2024 episode only, across all three pair constructions; the corresponding December 2024 and April 2025 figures, together with all raw outputs, daily tables, and plotting scripts, are available in the open-source companion repository (Furnes 2026). The reader is encouraged to consult the repository directly for the full grid of results across the three monthly episodes and the three pair constructions, and for any additional diagnostics not reproduced here.

The appendix is organised in three blocks. The first block is the three per-pair tearsheets in Section 8.1, one table per pair, stacking the three monthly episodes vertically so that return, risk, risk-adjusted performance, trading activity, and distributional diagnostics can be read side by side for the four strategies on a single page. The second block is a sequence of figure grids, with one plot type per page where possible: cross-pair cumulative-PnL panels on a symmetric-log axis in Section 8.2 (the same view of Figure 5 for the two pairs not shown in the body), rolling spread volatility in Section 8.3, per-bar period returns in Section 8.4, spread-return distributions for the two remaining pairs in Section 8.5, and per-day HMM regime dynamics in Section 8.6. The third block is the daily strategy summary tables of Section 8.7, one per pair, reporting every trading day in every episode at the round-trip level so that the bleed pattern of the Baseline and the gating behaviour of AR and MS can be inspected directly rather than only in aggregate. The appendix closes with a symbol dictionary (Section 8.8) intended as a lookup reference for the notation used in the methods sections.

8.1 Tearsheets

The five blocks in the tables below read as follows. The Return block reports the realised cumulative PnL in basis points over the episode (Total Return), its annualised counterpart (Annual Return, computed as bps $\times\, T_{\text{ann}}/T_{\text{episode}}$), and the best/worst monthly values; with only one month per episode, the best and worst columns coincide with the total and the "positive months" column is 0% or 100%. The Risk block reports annualised volatility σ_{ann} of (19), the maximum drawdown MDD and its duration in bars, the Ulcer index UI of (21), and the 95% Value-at-Risk and Conditional VaR of (20). The Risk-adj. block reports the Sharpe, Sortino, and Calmar ratios of (22), together with the trade-level Profit Factor (sum of winning trades divided by absolute sum of losing trades), Payoff Ratio (average win divided by average loss), and a Tail Ratio (the ratio of the 95th to the absolute 5th percentile of returns). The Trading block reports the number of round-trip trades, the win rate, the market exposure (fraction of bars with a non-zero position), and the average winning and losing trade in basis points. The Distr. block reports the per-bar skewness $\mathcal{S}$ and kurtosis $\mathcal{K}$ of (19), both of which are large in magnitude under the conditionally heteroskedastic, heavy-tailed return distribution discussed in Section 6.2.

Table 4: AUD/USD–NZD/USD strategy tearsheet by monthly episode.

Block	Month	Metric	BH	BL	AR	MS
Return	Aug. 2024	Total Return (bps)	-45.84	-966.47	-179.31	-206.06
		Annual Return (bps)	-577.34	-12172.56	-2258.32	-2595.27
		Best Month (bps)	-45.84	-966.47	-179.31	-206.06
		Worst Month (bps)	-45.84	-966.47	-179.31	-206.06
		Positive Months	0.00%	0.00%	0.00%	0.00%
	Dec. 2024	Total Return (bps)	-39.34	-776.45	-136.70	-158.02
		Annual Return (bps)	-495.49	-9779.28	-1721.69	-1990.25
		Best Month (bps)	-39.34	-776.45	-136.70	-158.02
		Worst Month (bps)	-39.34	-776.45	-136.70	-158.02
		Positive Months	0.00%	0.00%	0.00%	0.00%
	Apr. 2025	Total Return (bps)	155.23	-2521.35	-652.23	-731.32
		Annual Return (bps)	1955.13	-31756.03	-8214.73	-9210.85
		Best Month (bps)	155.23	-2521.35	-652.23	-731.32
		Worst Month (bps)	155.23	-2521.35	-652.23	-731.32
		Positive Months	100.00%	0.00%	0.00%	0.00%
Risk	Aug. 2024	Annual Volatility (bps)	554.99	360.20	120.07	181.98
		Max Drawdown (bps)	-171.64	-966.67	-179.31	-210.59
		Max DD Duration (bars)	2302	5330	4741	4999
		Ulcer Index	70.08	613.89	144.12	167.35
		VaR 95% (bps)	-3.29	-2.37	0.00	0.00
		CVaR 95% (bps)	-4.87	-4.08	-0.06	-0.10
	Dec. 2024	Annual Volatility (bps)	356.59	264.79	80.61	109.32
		Max Drawdown (bps)	-119.60	-778.78	-136.70	-162.72
		Max DD Duration (bars)	2698	5435	5321	5313
		Ulcer Index	49.36	412.81	68.77	80.60
		VaR 95% (bps)	-2.09	-1.85	0.00	0.00
		CVaR 95% (bps)	-3.09	-2.99	-0.04	-0.05
	Apr. 2025	Annual Volatility (bps)	743.57	541.55	225.04	252.92
		Max Drawdown (bps)	-198.20	-2521.66	-652.45	-737.07
		Max DD Duration (bars)	3452	10764	9796	10469
		Ulcer Index	64.99	1455.73	461.45	513.13
		VaR 95% (bps)	-3.21	-2.68	-0.64	-0.96
		CVaR 95% (bps)	-4.58	-4.44	-2.06	-2.32
Risk-adj.	Aug. 2024	Sharpe Ratio	-1.04	-33.79	-18.81	-14.26
		Sortino Ratio	-1.45	-24.89	-7.84	-7.03
		Calmar Ratio	3.36	12.59	12.59	12.32
		Profit Factor	0.99	0.51	0.47	0.59
		Payoff Ratio	1.00	0.81	0.86	0.93
		Tail Ratio	0.98	0.63	–	–
	Dec. 2024	Sharpe Ratio	-1.39	-36.93	-21.36	-18.21
		Sortino Ratio	-1.85	-23.03	-7.44	-5.80
		Calmar Ratio	4.14	12.56	12.59	12.23
		Profit Factor	0.98	0.44	0.38	0.44
		Payoff Ratio	0.97	0.67	0.55	0.62
		Tail Ratio	0.99	0.48	–	–
	Apr. 2025	Sharpe Ratio	2.63	-58.64	-36.50	-36.42
		Sortino Ratio	3.71	-46.67	-18.38	-19.30
		Calmar Ratio	9.86	12.59	12.59	12.50
		Profit Factor	1.02	0.46	0.44	0.47
		Payoff Ratio	1.03	0.73	0.71	0.70
		Tail Ratio	0.99	0.55	0.00	0.00
Trading	Aug. 2024	Number of Trades	0	193	39	45
		Win Rate	49.71%	38.82%	35.18%	38.95%
		Market Exposure	89.78%	31.25%	5.70%	7.82%
		Avg Win (bps)	1.60	1.56	1.45	1.83
		Avg Loss (bps)	-1.60	-1.93	-1.69	-1.97
	Dec. 2024	Number of Trades	0	186	38	41
		Win Rate	50.35%	39.65%	40.78%	41.39%
		Market Exposure	90.43%	27.03%	5.14%	6.03%
		Avg Win (bps)	0.99	1.04	0.72	0.89
		Avg Loss (bps)	-1.02	-1.55	-1.32	-1.45
	Apr. 2025	Number of Trades	0	434	140	148
		Win Rate	49.89%	38.96%	38.43%	39.86%
		Market Exposure	94.92%	35.23%	10.50%	12.07%
		Avg Win (bps)	1.49	1.47	1.18	1.23
		Avg Loss (bps)	-1.45	-2.02	-1.67	-1.74
Distr.	Aug. 2024	Skewness	0.56	-3.15	-2.65	0.73
		Kurtosis	11.73	61.41	45.38	67.15
	Dec. 2024	Skewness	0.06	-5.94	-4.67	-5.81
		Kurtosis	26.41	116.15	50.27	119.04
	Apr. 2025	Skewness	0.08	-2.06	-2.50	-2.45
		Kurtosis	4.87	28.09	34.11	34.01

Notes: BH = Buy-and-Hold; BL = baseline cointegration; AR = binary regime-gated; MS = continuous MS-AR threshold. A dash indicates an undefined or numerically unstable statistic.

Table 5: EUR/NOK–EUR/SEK strategy tearsheet by monthly episode.

Block	Month	Metric	BH	BL	AR	MS
Return	Aug. 2024	Total Return (bps)	-148.51	-6277.42	-1816.72	-2490.51
		Annual Return (bps)	-1870.45	-79062.97	-22881.30	-31367.54
		Best Month (bps)	-148.51	-6277.42	-1816.72	-2490.51
		Worst Month (bps)	-148.51	-6277.42	-1816.72	-2490.51
		Positive Months	0.00%	0.00%	0.00%	0.00%
	Dec. 2024	Total Return (bps)	60.58	-4473.83	-1592.84	-2070.78
		Annual Return (bps)	763.03	-56347.12	-20061.51	-26081.12
		Best Month (bps)	60.58	-4473.83	-1592.84	-2070.78
		Worst Month (bps)	60.58	-4473.83	-1592.84	-2070.78
		Positive Months	100.00%	0.00%	0.00%	0.00%
	Apr. 2025	Total Return (bps)	527.70	-19906.39	-8103.49	-9487.28
		Annual Return (bps)	6646.26	-250717.51	-102062.01	-119490.65
		Best Month (bps)	527.70	-19906.39	-8103.49	-9487.28
		Worst Month (bps)	527.70	-19906.39	-8103.49	-9487.28
		Positive Months	100.00%	0.00%	0.00%	0.00%
Risk	Aug. 2024	Annual Volatility (bps)	904.87	1226.06	501.14	836.51
		Max Drawdown (bps)	-459.55	-6277.42	-1816.72	-2492.83
		Max DD Duration (bars)	12611	14774	14236	14774
		Ulcer Index	316.62	3826.73	1139.70	1583.40
		VaR 95% (bps)	-3.27	-3.52	0.00	-0.12
		CVaR 95% (bps)	-4.99	-7.99	-0.14	-3.97
	Dec. 2024	Annual Volatility (bps)	634.41	813.82	343.78	532.97
		Max Drawdown (bps)	-212.66	-4473.83	-1592.84	-2070.78
		Max DD Duration (bars)	4166	10633	10500	10500
		Ulcer Index	72.69	2498.43	1004.40	1232.91
		VaR 95% (bps)	-2.58	-4.28	-0.39	-0.75
		CVaR 95% (bps)	-4.19	-6.97	-3.39	-4.50
	Apr. 2025	Annual Volatility (bps)	1508.79	1988.38	973.31	1143.00
		Max Drawdown (bps)	-474.38	-19906.39	-8103.49	-9487.28
		Max DD Duration (bars)	18798	26085	26085	26085
		Ulcer Index	294.93	11686.94	4533.58	5341.94
		VaR 95% (bps)	-3.56	-7.34	-0.68	-1.42
		CVaR 95% (bps)	-6.35	-11.93	-6.53	-7.71
Risk-adj.	Aug. 2024	Sharpe Ratio	-2.07	-64.49	-45.66	-37.50
		Sortino Ratio	-2.66	-35.21	-11.67	-9.99
		Calmar Ratio	4.07	12.59	12.59	12.58
		Profit Factor	0.99	0.32	0.11	0.16
		Payoff Ratio	0.98	0.49	0.18	0.24
		Tail Ratio	1.00	0.39	–	0.00
	Dec. 2024	Sharpe Ratio	1.20	-69.24	-58.36	-48.94
		Sortino Ratio	1.51	-40.33	-21.63	-17.54
		Calmar Ratio	3.59	12.59	12.59	12.59
		Profit Factor	1.01	0.27	0.14	0.18
		Payoff Ratio	1.03	0.42	0.21	0.26
		Tail Ratio	1.03	0.24	0.00	0.14
	Apr. 2025	Sharpe Ratio	4.41	-126.09	-104.86	-104.54
		Sortino Ratio	5.30	-76.34	-41.53	-44.25
		Calmar Ratio	14.01	12.59	12.59	12.59
		Profit Factor	1.03	0.21	0.08	0.11
		Payoff Ratio	1.03	0.35	0.14	0.19
		Tail Ratio	1.02	0.15	0.06	0.09
Trading	Aug. 2024	Number of Trades	0	606	137	158
		Win Rate	50.05%	39.65%	38.10%	39.15%
		Market Exposure	96.29%	34.84%	7.25%	9.22%
		Avg Win (bps)	1.38	1.45	0.54	0.86
		Avg Loss (bps)	-1.40	-2.97	-3.06	-3.55
	Dec. 2024	Number of Trades	0	435	158	181
		Win Rate	49.42%	39.48%	40.00%	40.44%
		Market Exposure	95.08%	35.19%	11.90%	14.64%
		Avg Win (bps)	1.13	1.14	0.50	0.71
		Avg Loss (bps)	-1.09	-2.71	-2.42	-2.70
	Apr. 2025	Number of Trades	0	1116	495	545
		Win Rate	50.08%	37.89%	36.63%	36.97%
		Market Exposure	97.90%	36.22%	14.85%	17.44%
		Avg Win (bps)	1.41	1.50	0.50	0.71
		Avg Loss (bps)	-1.38	-4.30	-3.59	-3.72
Distr.	Aug. 2024	Skewness	0.21	-13.00	-15.75	-22.55
		Kurtosis	21.37	308.63	415.35	740.30
	Dec. 2024	Skewness	-0.17	-10.48	-5.90	-15.88
		Kurtosis	12.07	251.63	40.05	458.00
	Apr. 2025	Skewness	0.48	-5.59	-6.44	-6.48
		Kurtosis	38.58	67.41	60.79	70.40

Notes: BH = Buy-and-Hold; BL = baseline cointegration; AR = binary regime-gated; MS = continuous MS-AR threshold. A dash indicates an undefined or numerically unstable statistic.

Table 6: GBP/USD–EUR/USD strategy tearsheet by monthly episode.

Block	Month	Metric	BH	BL	AR	MS
Return	Aug. 2024	Total Return (bps)	65.09	-396.06	-48.60	-60.14
		Annual Return (bps)	819.75	-4988.30	-612.05	-757.45
		Best Month (bps)	65.09	-396.06	-48.60	-60.14
		Worst Month (bps)	65.09	-396.06	-48.60	-60.14
		Positive Months	100.00%	0.00%	0.00%	0.00%
	Dec. 2024	Total Return (bps)	25.10	-730.39	-69.62	-75.81
		Annual Return (bps)	316.08	-9199.19	-876.90	-954.82
		Best Month (bps)	25.10	-730.39	-69.62	-75.81
		Worst Month (bps)	25.10	-730.39	-69.62	-75.81
		Positive Months	100.00%	0.00%	0.00%	0.00%
	Apr. 2025	Total Return (bps)	15.94	-1509.17	-350.61	-405.94
		Annual Return (bps)	200.78	-19007.67	-4415.92	-5112.73
		Best Month (bps)	15.94	-1509.17	-350.61	-405.94
		Worst Month (bps)	15.94	-1509.17	-350.61	-405.94
		Positive Months	100.00%	0.00%	0.00%	0.00%
Risk	Aug. 2024	Annual Volatility (bps)	497.41	289.35	103.71	130.86
		Max Drawdown (bps)	-171.91	-417.06	-67.12	-67.90
		Max DD Duration (bars)	5168	6985	6156	6985
		Ulcer Index	85.73	263.30	41.59	42.89
		VaR 95% (bps)	-2.56	-1.60	-0.00	-0.01
		CVaR 95% (bps)	-3.74	-2.77	-0.04	-1.08
	Dec. 2024	Annual Volatility (bps)	444.49	282.61	71.07	103.73
		Max Drawdown (bps)	-101.94	-738.06	-71.95	-79.82
		Max DD Duration (bars)	2879	8011	8011	8063
		Ulcer Index	51.72	401.64	47.07	55.06
		VaR 95% (bps)	-2.12	-1.48	0.00	0.00
		CVaR 95% (bps)	-3.22	-2.63	-0.03	-0.03
	Apr. 2025	Annual Volatility (bps)	644.21	416.02	135.39	174.93
		Max Drawdown (bps)	-332.36	-1525.73	-356.08	-420.51
		Max DD Duration (bars)	7736	14733	13115	14105
		Ulcer Index	146.04	906.32	212.03	249.97
		VaR 95% (bps)	-2.34	-1.65	-0.09	-0.27
		CVaR 95% (bps)	-3.36	-2.85	-0.94	-1.19
Risk-adj.	Aug. 2024	Sharpe Ratio	1.65	-17.24	-5.90	-5.79
		Sortino Ratio	2.26	-14.69	-2.52	-2.76
		Calmar Ratio	4.77	11.96	9.12	11.16
		Profit Factor	1.02	0.76	0.83	0.84
		Payoff Ratio	1.00	1.04	1.03	1.04
		Tail Ratio	0.99	0.85	–	0.00
	Dec. 2024	Sharpe Ratio	0.71	-32.55	-12.34	-9.20
		Sortino Ratio	0.95	-25.99	-4.99	-3.39
		Calmar Ratio	3.10	12.46	12.19	11.96
		Profit Factor	1.01	0.61	0.65	0.72
		Payoff Ratio	1.00	0.93	1.03	1.07
		Tail Ratio	1.02	0.71	–	–
	Apr. 2025	Sharpe Ratio	0.31	-45.69	-32.62	-29.23
		Sortino Ratio	0.45	-37.03	-14.91	-13.20
		Calmar Ratio	0.60	12.46	12.40	12.16
		Profit Factor	1.00	0.60	0.50	0.55
		Payoff Ratio	1.01	0.92	0.86	0.89
		Tail Ratio	1.01	0.68	0.00	0.00
Trading	Aug. 2024	Number of Trades	0	281	79	83
		Win Rate	50.48%	42.26%	44.49%	44.87%
		Market Exposure	92.27%	33.50%	7.39%	9.18%
		Avg Win (bps)	1.23	1.26	1.00	1.10
		Avg Loss (bps)	-1.24	-1.21	-0.96	-1.06
	Dec. 2024	Number of Trades	0	325	58	63
		Win Rate	50.28%	39.48%	38.66%	40.29%
		Market Exposure	93.54%	35.31%	5.86%	6.84%
		Avg Win (bps)	1.01	1.01	0.70	0.86
		Avg Loss (bps)	-1.01	-1.08	-0.68	-0.81
	Apr. 2025	Number of Trades	0	590	154	161
		Win Rate	49.78%	39.64%	36.73%	38.33%
		Market Exposure	96.29%	35.64%	8.60%	10.13%
		Avg Win (bps)	1.09	1.09	0.75	0.87
		Avg Loss (bps)	-1.08	-1.19	-0.87	-0.98
Distr.	Aug. 2024	Skewness	-0.01	-0.29	-0.49	0.18
		Kurtosis	11.41	12.48	52.25	54.06
	Dec. 2024	Skewness	-0.32	-1.16	0.90	-1.03
		Kurtosis	9.58	25.35	122.03	334.80
	Apr. 2025	Skewness	0.25	-1.15	-1.83	-2.87
		Kurtosis	4.75	20.62	37.78	66.55

Notes: BH = Buy-and-Hold; BL = baseline cointegration; AR = binary regime-gated; MS = continuous MS-AR threshold. A dash indicates an undefined or numerically unstable statistic.

8.2 Three-month cumulative-PnL panels (symlog)

Symlog-scaled cumulative-PnL panels stacking the four strategies (Buy & Hold, Baseline, AR, MS-AR) across the three monthly episodes for each of the three pair constructions.

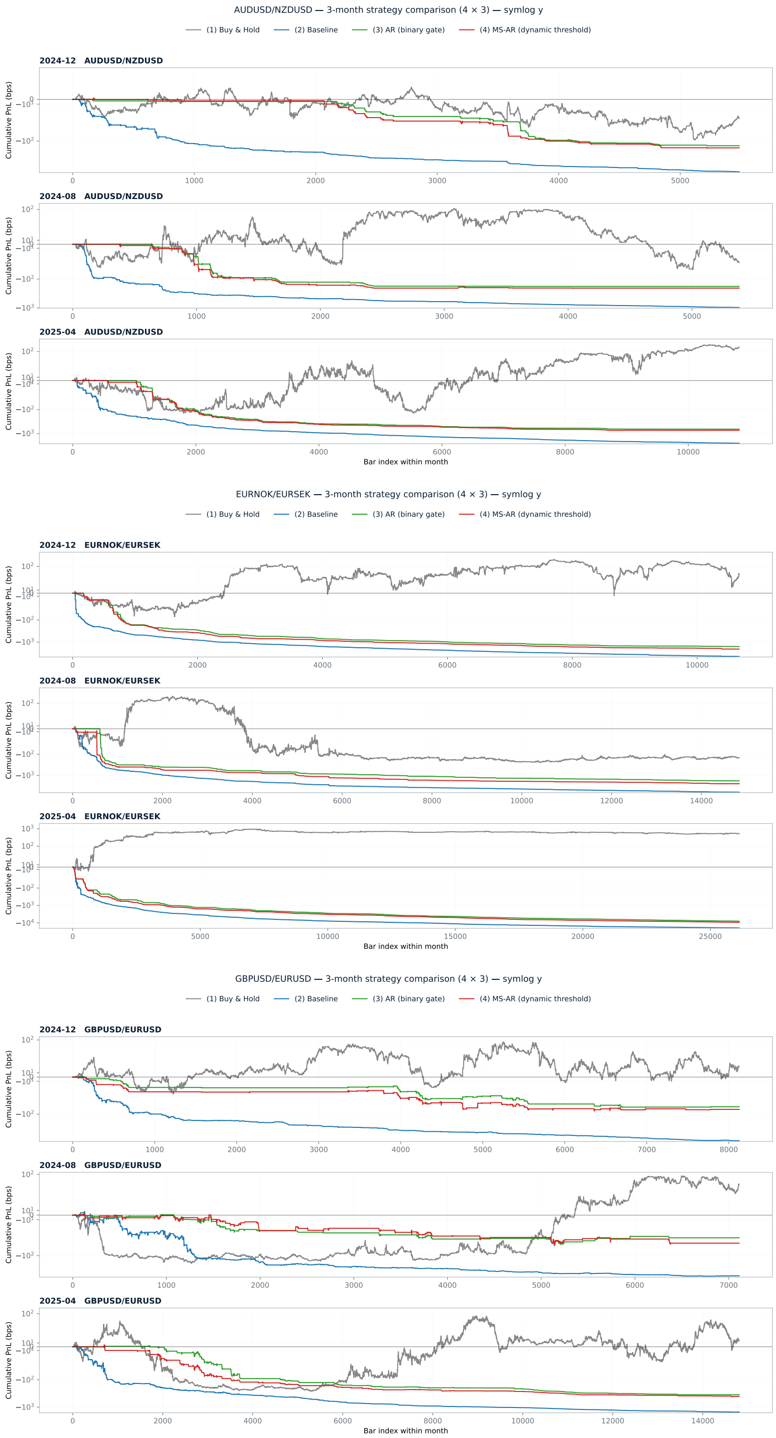

Figure 6: Three-month strategy comparison on a symmetric-log y-axis. Top: AUD/USD–NZD/USD; middle: EUR/NOK–EUR/SEK; bottom: GBP/USD–EUR/USD. Each panel stacks the four cumulative-PnL curves across the three monthly episodes (August 2024, December 2024, April 2025).

8.3 Rolling volatility

Rolling realised volatility of the spread return ΔS_t, for the August 2024 episode of each pair. The remaining months are available in the companion repository (Furnes 2026).

8.3.1 AUD/USD–NZD/USD

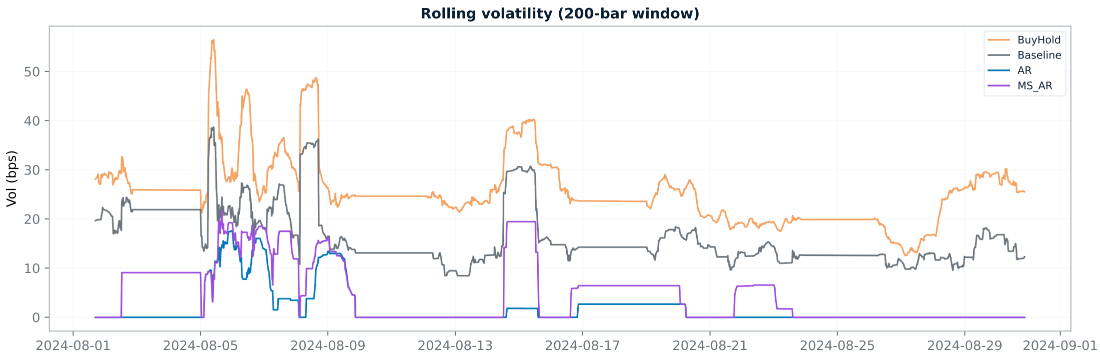

Figure 7: AUD/USD–NZD/USD, August 2024: rolling spread volatility.

8.3.2 EUR/NOK–EUR/SEK

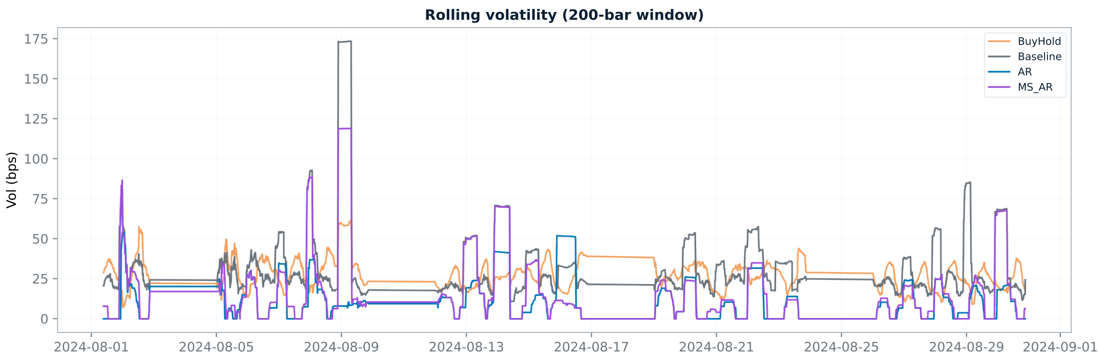

Figure 8: EUR/NOK–EUR/SEK, August 2024: rolling spread volatility.

8.3.3 GBP/USD–EUR/USD

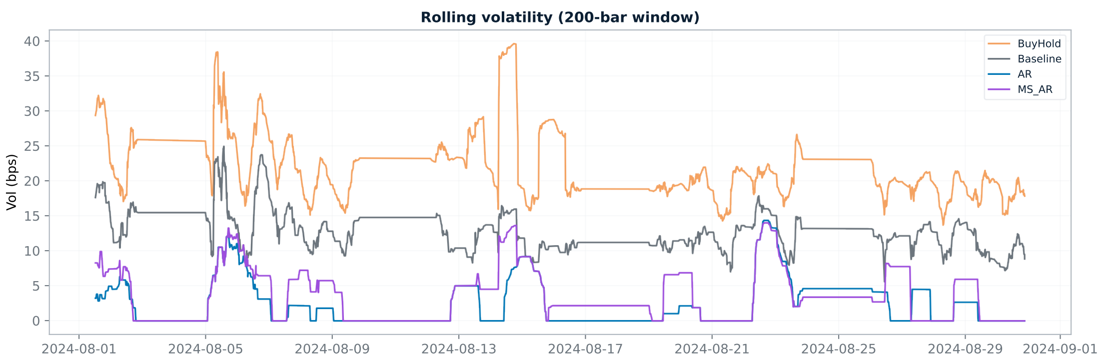

Figure 9: GBP/USD–EUR/USD, August 2024: rolling spread volatility.

8.4 Period returns

Per-bar period returns for the August 2024 episode of each pair. The remaining months are available in the companion repository (Furnes 2026).

8.4.1 AUD/USD–NZD/USD

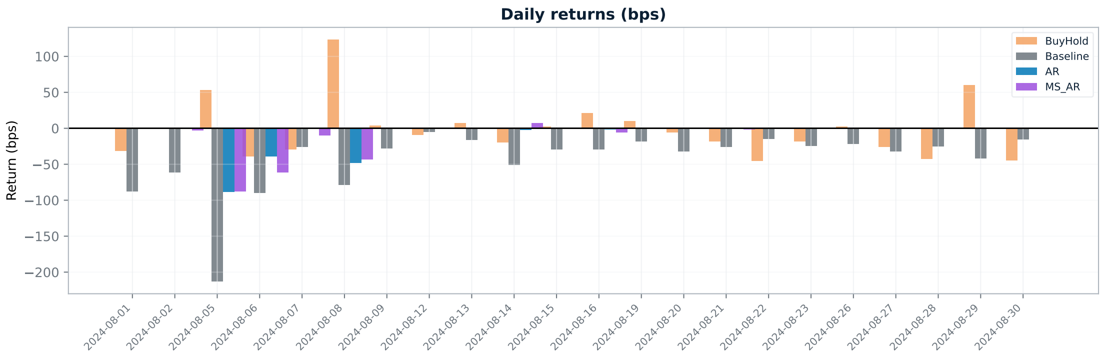

Figure 10: AUD/USD–NZD/USD, August 2024: period returns.

8.4.2 EUR/NOK–EUR/SEK

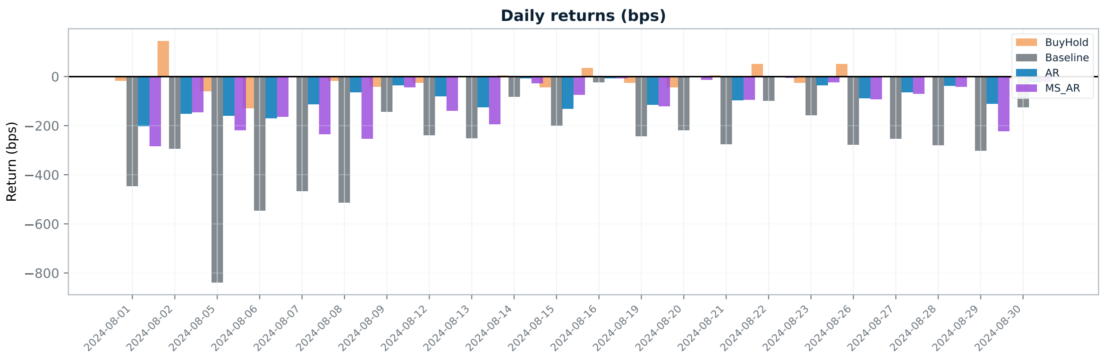

Figure 11: EUR/NOK–EUR/SEK, August 2024: period returns.

8.4.3 GBP/USD–EUR/USD

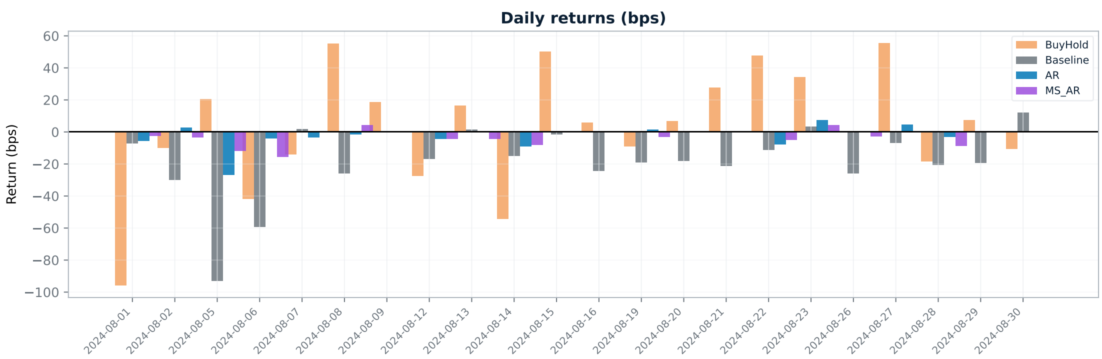

Figure 12: GBP/USD–EUR/USD, August 2024: period returns.

8.5 Spread-return distributions

Per-bar spread-return distributions for the two pairs not shown in the body. Each pair occupies two rows (linear y-axis on top, log y-axis below); columns are August 2024, December 2024, and April 2025.

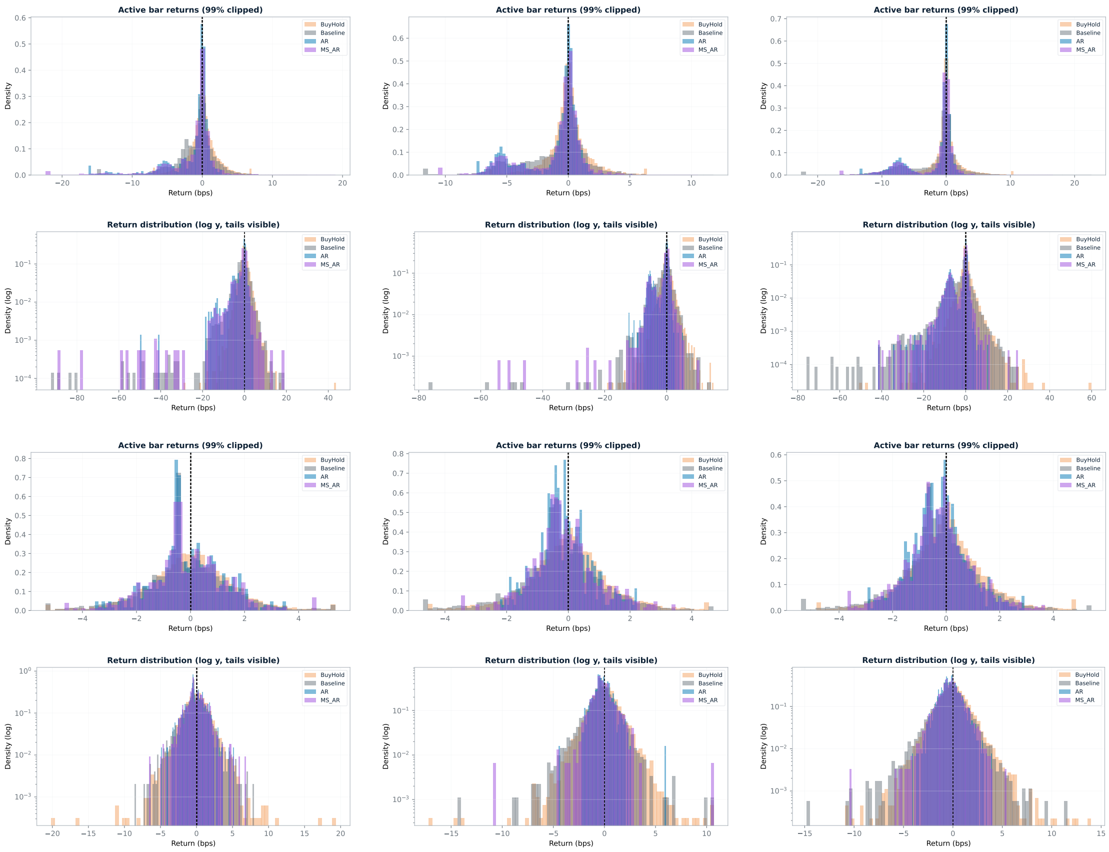

Figure 13: Spread-return distributions for EUR/NOK–EUR/SEK (rows 1–2) and GBP/USD–EUR/USD (rows 3–4). Each pair: top row linear y-axis, bottom row log y-axis. Columns: August 2024 (left), December 2024 (centre), April 2025 (right).

8.6 Markov regime dynamics

Per-day HMM fits showing regime-specific innovation variances $\sigma^{(k)}$, persistence probabilities p_{ii}, and regime means $\mu^{(k)}$, for the August 2024 episode of each pair. The remaining months are available in the companion repository (Furnes 2026).

8.6.1 AUD/USD–NZD/USD

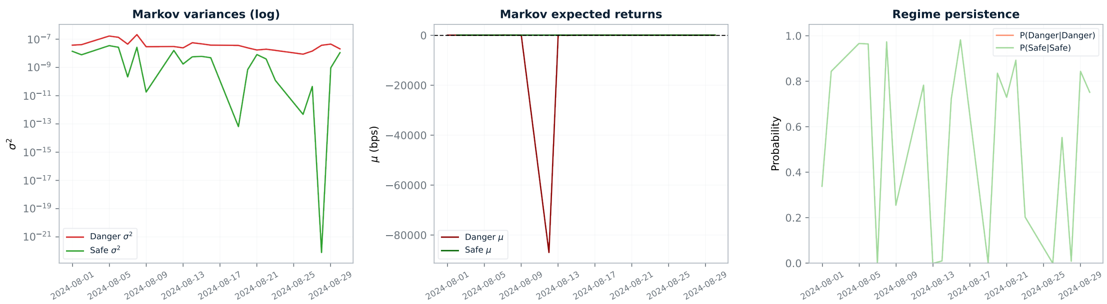

Figure 14: AUD/USD–NZD/USD, August 2024: HMM regime dynamics.

8.6.2 EUR/NOK–EUR/SEK

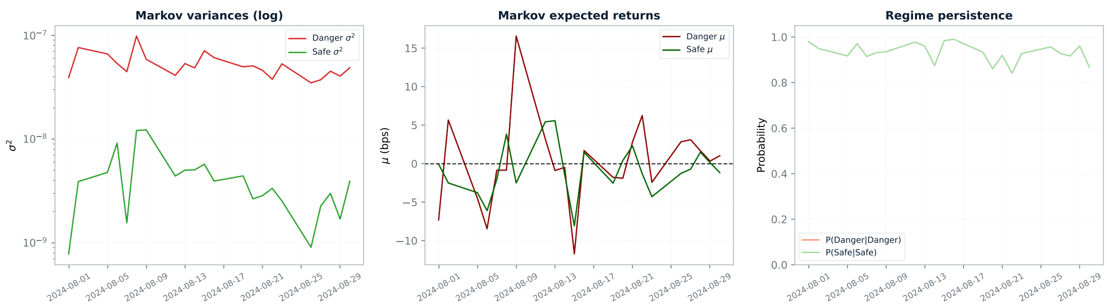

Figure 15: EUR/NOK–EUR/SEK, August 2024: HMM regime dynamics.

8.6.3 GBP/USD–EUR/USD

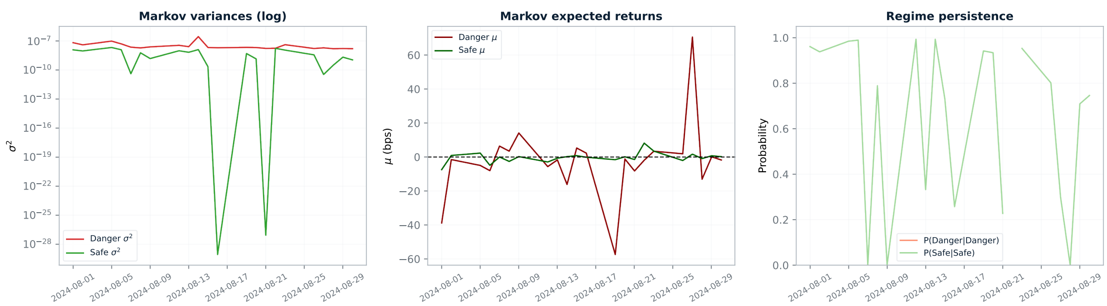

Figure 16: GBP/USD–EUR/USD, August 2024: HMM regime dynamics.

8.7 Daily strategy summaries (raw ledgers)

The three tables that follow are the raw daily ledgers underlying every aggregate number reported in the body and the tearsheets, with one row per trading day, one block per monthly episode, and the same four strategy columns (BH, BL, AR, MS) used throughout the thesis. They are deliberately compact, so the abbreviations are dense; once decoded, every aggregate number in Tables 2–3 and in the tearsheets above can be reconstructed by summing or averaging the corresponding daily entries.

Each cell follows the format `POSITION, ACTIVITY; PNL`, where POSITION describes the position held at the start of the day (L for long, S for short, F/M for flat-or-mixed within the day for the active strategies), ACTIVITY reports the trading activity (`NF` for no flips and a flat position the entire day, `n RT` for n completed round-trip trades), and PNL is the day's profit-and-loss in basis points, signed and net of the half-spread plus 0.05 bps commission cost model. The BH column always shows `L, NF` because Buy & Hold simply holds a unit long-spread position throughout, so its daily PnL is the realised ΔS_t aggregated over the day. The BL column shows non-trivial trading activity on every day (the rolling z-score rule flips 5 to 40 times per day depending on volatility), while the AR and MS columns frequently show `NF` because the HMM gate G_t of (16) or the regime-weighted threshold of (18) suppresses entries in the drifting regime. The bold `Total` row at the end of each monthly block is the sum of the corresponding daily PnL column for that strategy in that month, and matches the total-return row of the tearsheets up to floating-point rounding.

Table 7: Daily AUD/USD–NZD/USD strategy summaries.

Date	BH	BL	AR	MS
August 2024				
08-01	L, NF; -31.34	F/M, 11RT; -87.51	F/M, NF; +0.00	F/M, NF; +0.00
08-02	L, NF; +0.48	F/M, 13RT; -61.19	F/M, NF; +0.00	F/M, 1RT; -3.14
08-05	L, NF; +53.17	F/M, 25RT; -213.20	F/M, 16RT; -88.65	F/M, 14RT; -87.77
08-06	L, NF; -39.24	F/M, 19RT; -90.09	F/M, 11RT; -38.79	F/M, 12RT; -61.67
08-07	L, NF; -29.49	F/M, 12RT; -25.74	F/M, 2RT; -0.34	F/M, 2RT; -10.24
08-08	L, NF; +123.49	F/M, 10RT; -78.58	F/M, 7RT; -47.92	F/M, 9RT; -42.99
08-09	L, NF; +4.02	F/M, 4RT; -28.14	F/M, NF; +0.00	F/M, NF; +0.00
08-12	L, NF; -9.05	F/M, 3RT; -4.94	F/M, NF; +0.00	F/M, NF; +0.00
08-13	L, NF; +7.00	F/M, 5RT; -16.18	F/M, NF; +0.00	F/M, NF; +0.00
08-14	L, NF; -19.86	F/M, 8RT; -50.83	F/M, 1RT; -2.11	F/M, 2RT; +7.35
08-15	L, NF; +2.26	F/M, 7RT; -29.29	F/M, NF; +0.00	F/M, NF; +0.00
08-16	L, NF; +21.36	F/M, 3RT; -29.23	F/M, 1RT; -1.48	F/M, 2RT; -6.00
08-19	L, NF; +9.80	F/M, 5RT; -18.18	F/M, NF; +0.00	F/M, NF; +0.00
08-20	L, NF; -5.42	F/M, 6RT; -32.01	F/M, NF; +0.00	F/M, NF; +0.00
08-21	L, NF; -18.57	F/M, 4RT; -26.17	F/M, NF; +0.00	F/M, 1RT; -1.36
08-22	L, NF; -45.36	F/M, 9RT; -14.86	F/M, NF; +0.00	F/M, 1RT; -0.25
08-23	L, NF; -18.31	F/M, 5RT; -24.71	F/M, NF; +0.00	F/M, NF; +0.00
08-26	L, NF; +2.29	F/M, 9RT; -21.43	F/M, NF; +0.00	F/M, NF; +0.00
08-27	L, NF; -25.98	F/M, 6RT; -31.83	F/M, NF; +0.00	F/M, NF; +0.00
08-28	L, NF; -42.90	F/M, 7RT; -24.97	F/M, NF; +0.00	F/M, NF; +0.00
08-29	L, NF; +60.27	F/M, 8RT; -41.63	F/M, NF; +0.00	F/M, NF; +0.00
08-30	L, NF; -44.46	F/M, 5RT; -15.78	F/M, NF; +0.00	F/M, NF; +0.00
Total	**-45.80**	**-966.50**	**-179.29**	**-206.08**
December 2024				
12-02	L, NF; -23.45	F/M, 13RT; -53.50	F/M, 1RT; -3.17	F/M, 1RT; -0.21
12-03	L, NF; +14.45	F/M, 6RT; -8.46	F/M, 1RT; -1.22	F/M, 1RT; -1.22
12-04	L, NF; +10.09	F/M, 12RT; -55.87	F/M, NF; +0.00	F/M, 1RT; -1.64
12-05	L, NF; -13.59	F/M, 7RT; -23.94	F/M, NF; +0.00	F/M, NF; +0.00
12-06	L, NF; -6.79	F/M, 10RT; -41.15	F/M, NF; +0.00	F/M, NF; +0.00
12-09	L, NF; +22.70	F/M, 6RT; -21.88	F/M, NF; +0.00	F/M, NF; +0.00
12-10	L, NF; -11.36	F/M, 6RT; -6.40	F/M, NF; +0.00	F/M, 1RT; +0.32
12-11	L, NF; +7.31	F/M, 17RT; -63.10	F/M, 5RT; -13.85	F/M, 6RT; -26.97
12-12	L, NF; +2.42	F/M, 12RT; -45.77	F/M, 5RT; -17.46	F/M, 5RT; -15.63
12-13	L, NF; +5.05	F/M, 8RT; -29.26	F/M, NF; +0.00	F/M, NF; +0.00
12-16	L, NF; -22.37	F/M, 7RT; -15.01	F/M, NF; +0.00	F/M, NF; +0.00
12-17	L, NF; -8.15	F/M, 5RT; -17.33	F/M, 1RT; -3.57	F/M, 1RT; -1.71
12-18	L, NF; -6.43	F/M, 6RT; -73.73	F/M, 5RT; -7.48	F/M, 6RT; -29.82
12-19	L, NF; +16.30	F/M, 12RT; -68.77	F/M, 7RT; -48.28	F/M, 6RT; -23.91
12-20	L, NF; -23.04	F/M, 11RT; -42.48	F/M, 6RT; -17.80	F/M, 5RT; -20.30
12-23	L, NF; -9.43	F/M, 8RT; -28.51	F/M, 1RT; -2.02	F/M, 1RT; -2.02
12-24	L, NF; -5.02	F/M, 4RT; -30.63	F/M, NF; +0.00	F/M, NF; +0.00
12-26	L, NF; +12.32	F/M, 7RT; -45.35	F/M, 4RT; -16.94	F/M, 5RT; -33.32
12-27	L, NF; -16.39	F/M, 7RT; -36.77	F/M, NF; +0.00	F/M, NF; +0.00
12-30	L, NF; -14.99	F/M, 6RT; -42.23	F/M, 1RT; -4.92	F/M, 1RT; -1.59
12-31	L, NF; +31.04	F/M, 9RT; -26.34	F/M, NF; +0.00	F/M, NF; +0.00
Total	**-39.30**	**-776.48**	**-136.71**	**-158.00**
April 2025				
04-01	L, NF; +0.13	F/M, 6RT; -38.05	F/M, NF; +0.00	F/M, NF; +0.00
04-02	L, NF; -21.94	F/M, 14RT; -60.07	F/M, NF; +0.00	F/M, 1RT; +0.00
04-03	L, NF; -6.77	F/M, 17RT; -91.15	F/M, NF; +0.00	F/M, 1RT; -5.26
04-04	L, NF; -86.88	F/M, 34RT; -165.76	F/M, 17RT; -93.99	F/M, 15RT; -81.47
04-07	L, NF; +52.62	F/M, 44RT; -328.86	F/M, 25RT; -155.86	F/M, 28RT; -192.20
04-08	L, NF; +19.78	F/M, 26RT; -116.89	F/M, 18RT; -77.60	F/M, 17RT; -74.08
04-09	L, NF; +52.03	F/M, 38RT; -166.21	F/M, 18RT; -74.35	F/M, 19RT; -85.66
04-10	L, NF; -68.00	F/M, 29RT; -227.84	F/M, 8RT; -36.89	F/M, 11RT; -51.01
04-11	L, NF; +52.30	F/M, 38RT; -203.31	F/M, 18RT; -75.91	F/M, 14RT; -51.31
04-14	L, NF; +17.40	F/M, 18RT; -82.49	F/M, 6RT; -29.42	F/M, 7RT; -27.67
04-15	L, NF; -6.36	F/M, 12RT; -66.81	F/M, NF; +0.00	F/M, 2RT; -18.00
04-16	L, NF; +20.97	F/M, 16RT; -63.10	F/M, 14RT; -43.28	F/M, 14RT; -57.86
04-17	L, NF; -5.12	F/M, 9RT; -83.37	F/M, 2RT; -6.23	F/M, 3RT; -13.96
04-18	L, NF; +55.04	F/M, 48RT; -403.09	F/M, 7RT; -52.00	F/M, 10RT; -75.44
04-21	L, NF; +2.23	F/M, 10RT; -40.86	F/M, 1RT; -2.54	F/M, NF; +5.08
04-22	L, NF; -7.56	F/M, 14RT; -77.44	F/M, 1RT; +1.58	F/M, NF; +0.00
04-23	L, NF; +31.92	F/M, 13RT; -48.66	F/M, NF; +0.00	F/M, NF; +0.00
04-24	L, NF; +4.48	F/M, 13RT; -50.70	F/M, NF; +0.00	F/M, NF; +0.00
04-25	L, NF; +38.03	F/M, 9RT; -71.88	F/M, NF; +0.00	F/M, NF; +0.00
04-28	L, NF; +41.98	F/M, 7RT; -52.53	F/M, 1RT; -2.49	F/M, 1RT; -2.49
04-29	L, NF; -26.78	F/M, 6RT; -33.65	F/M, 1RT; -2.04	F/M, 1RT; +1.13
04-30	L, NF; -4.29	F/M, 7RT; -48.66	F/M, 1RT; -1.21	F/M, 1RT; -1.13
Total	**+155.20**	**-2521.40**	**-652.20**	**-731.30**

Notes: BH = Buy-and-Hold; BL = Baseline; AR = binary AR regime gate; MS = MS-AR continuous-threshold rule. L = long; F/M = flat/mixed; NF = no flips; RT = approximate round-trip(s). Values are daily PnL in basis points.

Table 8: Daily EUR/NOK–EUR/SEK strategy summaries.

Date	BH	BL	AR	MS
August 2024				
08-01	L, NF; -18.20	F/M, 29RT; -446.12	F/M, 7RT; -201.46	F/M, 8RT; -284.68
08-02	L, NF; +145.51	F/M, 36RT; -293.29	F/M, 12RT; -150.88	F/M, 13RT; -145.72
08-05	L, NF; -59.15	F/M, 65RT; -839.32	F/M, 13RT; -159.51	F/M, 16RT; -218.97
08-06	L, NF; -129.52	F/M, 45RT; -546.57	F/M, 10RT; -168.80	F/M, 11RT; -163.92
08-07	L, NF; -0.51	F/M, 37RT; -466.82	F/M, 6RT; -112.94	F/M, 8RT; -235.61
08-08	L, NF; -16.30	F/M, 24RT; -513.76	F/M, 7RT; -63.19	F/M, 9RT; -253.59
08-09	L, NF; -40.90	F/M, 21RT; -143.92	F/M, 7RT; -36.39	F/M, 7RT; -44.56
08-12	L, NF; -24.75	F/M, 21RT; -238.40	F/M, 7RT; -81.19	F/M, 8RT; -138.30
08-13	L, NF; -3.94	F/M, 23RT; -251.14	F/M, 10RT; -124.40	F/M, 11RT; -195.03
08-14	L, NF; +4.43	F/M, 12RT; -82.69	F/M, 1RT; -7.49	F/M, 2RT; -27.24
08-15	L, NF; -43.86	F/M, 24RT; -200.67	F/M, 11RT; -130.40	F/M, 10RT; -73.57
08-16	L, NF; +34.20	F/M, 15RT; -22.71	F/M, 1RT; -7.87	F/M, 1RT; -7.87
08-19	L, NF; -26.30	F/M, 27RT; -243.01	F/M, 8RT; -115.31	F/M, 7RT; -120.18
08-20	L, NF; -43.63	F/M, 27RT; -218.99	F/M, NF; +0.00	F/M, 3RT; -12.75
08-21	L, NF; +3.32	F/M, 27RT; -276.27	F/M, 5RT; -96.67	F/M, 5RT; -94.16
08-22	L, NF; +50.77	F/M, 12RT; -97.74	F/M, NF; +0.00	F/M, 1RT; -4.31
08-23	L, NF; -25.10	F/M, 18RT; -157.05	F/M, 4RT; -36.26	F/M, 3RT; -24.21
08-26	L, NF; +51.45	F/M, 32RT; -277.93	F/M, 8RT; -88.90	F/M, 9RT; -92.07
08-27	L, NF; +0.63	F/M, 27RT; -253.49	F/M, 4RT; -63.65	F/M, 5RT; -69.94
08-28	L, NF; -1.52	F/M, 25RT; -279.21	F/M, 4RT; -37.30	F/M, 4RT; -40.84
08-29	L, NF; +2.24	F/M, 26RT; -302.57	F/M, 8RT; -111.44	F/M, 11RT; -222.58
08-30	L, NF; -7.39	F/M, 27RT; -125.76	F/M, 3RT; -22.69	F/M, 5RT; -20.42
Total	**-148.52**	**-6277.43**	**-1816.74**	**-2490.52**
December 2024				
12-02	L, NF; -39.79	F/M, 29RT; -313.72	F/M, 6RT; -61.49	F/M, 5RT; -68.03
12-03	L, NF; -11.34	F/M, 27RT; -271.73	F/M, 17RT; -163.55	F/M, 19RT; -218.26
12-04	L, NF; +26.20	F/M, 27RT; -256.08	F/M, 9RT; -88.75	F/M, 12RT; -118.08
12-05	L, NF; +94.66	F/M, 27RT; -275.13	F/M, 15RT; -171.88	F/M, 18RT; -237.77
12-06	L, NF; +34.66	F/M, 25RT; -164.41	F/M, 6RT; -66.95	F/M, 6RT; -63.32
12-09	L, NF; -60.71	F/M, 21RT; -196.73	F/M, 7RT; -64.80	F/M, 7RT; -59.77
12-10	L, NF; +7.11	F/M, 22RT; -224.43	F/M, 12RT; -140.26	F/M, 12RT; -145.91
12-11	L, NF; +8.96	F/M, 17RT; -185.45	F/M, 6RT; -52.74	F/M, 9RT; -102.84
12-12	L, NF; -22.14	F/M, 22RT; -220.72	F/M, 8RT; -78.50	F/M, 8RT; -77.38
12-13	L, NF; +20.32	F/M, 21RT; -174.68	F/M, 6RT; -51.65	F/M, 7RT; -57.23
12-16	L, NF; +12.13	F/M, 26RT; -266.25	F/M, 14RT; -131.47	F/M, 15RT; -135.04
12-17	L, NF; +11.70	F/M, 15RT; -182.22	F/M, 6RT; -46.96	F/M, 9RT; -69.45
12-18	L, NF; +19.65	F/M, 17RT; -182.89	F/M, 6RT; -62.83	F/M, 6RT; -95.34
12-19	L, NF; +85.01	F/M, 20RT; -163.33	F/M, 8RT; -95.65	F/M, 10RT; -113.75
12-20	L, NF; -122.08	F/M, 26RT; -308.33	F/M, 11RT; -123.09	F/M, 12RT; -152.12
12-23	L, NF; +2.03	F/M, 25RT; -225.71	F/M, 8RT; -97.23	F/M, 8RT; -97.95
12-24	L, NF; +14.16	F/M, 8RT; -248.71	F/M, 1RT; -9.50	F/M, 1RT; -9.50
12-26	L, NF; +95.74	F/M, 14RT; -168.21	F/M, NF; +0.00	F/M, 1RT; -33.15
12-27	L, NF; -54.01	F/M, 16RT; -158.75	F/M, 2RT; -13.16	F/M, 3RT; -28.17
12-30	L, NF; -41.37	F/M, 16RT; -202.59	F/M, 8RT; -65.22	F/M, 10RT; -175.01
12-31	L, NF; -20.30	F/M, 9RT; -83.76	F/M, 1RT; -7.18	F/M, 2RT; -12.69
Total	**+60.59**	**-4473.83**	**-1592.86**	**-2070.76**
April 2025				
04-01	L, NF; +1.87	F/M, 18RT; -240.30	F/M, 5RT; -49.26	F/M, 5RT; -49.14
04-02	L, NF; +88.21	F/M, 17RT; -241.28	F/M, 9RT; -79.96	F/M, 12RT; -152.18
04-03	L, NF; +97.48	F/M, 37RT; -495.85	F/M, 15RT; -182.36	F/M, 17RT; -214.19
04-04	L, NF; +180.78	F/M, 29RT; -446.44	F/M, 15RT; -171.06	F/M, 17RT; -256.78
04-07	L, NF; +231.98	F/M, 44RT; -1213.70	F/M, 16RT; -410.56	F/M, 15RT; -404.87
04-08	L, NF; +21.78	F/M, 28RT; -454.75	F/M, 11RT; -188.16	F/M, 12RT; -233.40
04-09	L, NF; -56.74	F/M, 50RT; -1071.52	F/M, 12RT; -230.54	F/M, 19RT; -340.36
04-10	L, NF; +361.99	F/M, 58RT; -1203.87	F/M, 23RT; -384.66	F/M, 20RT; -378.45
04-11	L, NF; -190.56	F/M, 53RT; -1108.65	F/M, 21RT; -486.68	F/M, 21RT; -503.42
04-14	L, NF; -119.72	F/M, 57RT; -1186.25	F/M, 24RT; -399.04	F/M, 28RT; -526.91
04-15	L, NF; +49.63	F/M, 64RT; -826.57	F/M, 31RT; -408.54	F/M, 28RT; -373.06
04-16	L, NF; +44.51	F/M, 56RT; -930.52	F/M, 23RT; -449.51	F/M, 30RT; -543.34
04-17	L, NF; -82.92	F/M, 62RT; -836.64	F/M, 38RT; -542.80	F/M, 37RT; -569.53
04-18	L, NF; +33.90	F/M, 84RT; -1957.30	F/M, 27RT; -560.49	F/M, 37RT; -815.25
04-21	L, NF; +1.90	F/M, 71RT; -1346.75	F/M, 43RT; -689.11	F/M, 49RT; -922.87
04-22	L, NF; -87.83	F/M, 62RT; -1181.46	F/M, 24RT; -383.63	F/M, 29RT; -472.88
04-23	L, NF; +45.25	F/M, 83RT; -1573.22	F/M, 33RT; -493.71	F/M, 36RT; -622.10
04-24	L, NF; -21.95	F/M, 54RT; -780.65	F/M, 28RT; -446.71	F/M, 29RT; -439.88
04-25	L, NF; -21.48	F/M, 42RT; -591.91	F/M, 27RT; -410.07	F/M, 26RT; -369.03
04-28	L, NF; -65.16	F/M, 35RT; -565.56	F/M, 17RT; -256.15	F/M, 19RT; -334.78
04-29	L, NF; +13.06	F/M, 51RT; -913.39	F/M, 27RT; -497.57	F/M, 27RT; -489.51
04-30	L, NF; +1.70	F/M, 54RT; -739.84	F/M, 25RT; -382.90	F/M, 30RT; -475.34
Total	**+527.68**	**-19906.42**	**-8103.47**	**-9487.27**

Notes: BH = Buy-and-Hold; BL = Baseline; AR = binary AR regime gate; MS = MS-AR continuous-threshold rule. L = long; F/M = flat/mixed; NF = no flips; RT = approximate round-trip(s). Values are daily PnL in basis points.

Table 9: Daily GBP/USD–EUR/USD strategy summaries.

Date	BH	BL	AR	MS
August 2024				
08-01	L, NF; -95.81	F/M, 16RT; -7.01	F/M, 7RT; -5.37	F/M, 7RT; -2.36
08-02	L, NF; -9.79	F/M, 18RT; -29.83	F/M, 5RT; +2.74	F/M, 5RT; -3.24
08-05	L, NF; +20.45	F/M, 38RT; -93.04	F/M, 24RT; -26.81	F/M, 23RT; -11.86
08-06	L, NF; -41.77	F/M, 20RT; -59.23	F/M, 8RT; -4.02	F/M, 9RT; -15.54
08-07	L, NF; -13.88	F/M, 14RT; +1.63	F/M, 2RT; -3.30	F/M, 3RT; +0.35
08-08	L, NF; +55.12	F/M, 13RT; -25.94	F/M, 1RT; -1.33	F/M, 2RT; +4.34
08-09	L, NF; +18.42	F/M, 6RT; -0.35	F/M, NF; +0.00	F/M, NF; +0.00
08-12	L, NF; -27.43	F/M, 5RT; -16.67	F/M, 2RT; -4.26	F/M, 2RT; -4.26
08-13	L, NF; +16.26	F/M, 11RT; +1.41	F/M, NF; +0.00	F/M, 1RT; -4.32
08-14	L, NF; -54.31	F/M, 11RT; -14.84	F/M, 8RT; -8.98	F/M, 9RT; -8.11
08-15	L, NF; +50.15	F/M, 7RT; -1.52	F/M, NF; +0.00	F/M, 1RT; +0.18
08-16	L, NF; +5.67	F/M, 5RT; -24.36	F/M, NF; +0.00	F/M, NF; +0.00
08-19	L, NF; -8.97	F/M, 8RT; -19.09	F/M, 2RT; +1.47	F/M, 2RT; -3.12
08-20	L, NF; +6.62	F/M, 11RT; -18.05	F/M, NF; +0.00	F/M, NF; +0.00
08-21	L, NF; +27.71	F/M, 13RT; -21.26	F/M, NF; +0.00	F/M, NF; +0.00
08-22	L, NF; +47.55	F/M, 14RT; -11.02	F/M, 13RT; -7.55	F/M, 13RT; -4.86
08-23	L, NF; +34.32	F/M, 12RT; +3.20	F/M, 4RT; +7.36	F/M, 3RT; +4.26
08-26	L, NF; +0.52	F/M, 8RT; -25.83	F/M, NF; +0.00	F/M, 1RT; -2.85
08-27	L, NF; +55.37	F/M, 11RT; -6.69	F/M, 1RT; +4.48	F/M, NF; +0.00
08-28	L, NF; -18.20	F/M, 12RT; -20.50	F/M, 1RT; -3.03	F/M, 1RT; -8.76
08-29	L, NF; +7.45	F/M, 12RT; -19.10	F/M, NF; +0.00	F/M, NF; +0.00
08-30	L, NF; -10.38	F/M, 13RT; +12.00	F/M, NF; +0.00	F/M, NF; +0.00
Total	**+65.07**	**-396.09**	**-48.60**	**-60.15**
December 2024				
12-02	L, NF; +3.19	F/M, 27RT; -58.38	F/M, 8RT; -8.08	F/M, 8RT; -18.27
12-03	L, NF; -12.52	F/M, 18RT; -40.69	F/M, 6RT; -16.33	F/M, 6RT; -16.33
12-04	L, NF; +15.83	F/M, 21RT; -61.39	F/M, NF; +0.00	F/M, NF; +0.00
12-05	L, NF; -3.05	F/M, 16RT; -2.69	F/M, 1RT; -0.72	F/M, 1RT; -0.72
12-06	L, NF; +13.69	F/M, 17RT; -26.09	F/M, NF; +0.00	F/M, 1RT; +0.53
12-09	L, NF; -5.84	F/M, 15RT; -37.29	F/M, NF; +0.00	F/M, NF; +0.00
12-10	L, NF; +60.00	F/M, 15RT; -25.24	F/M, NF; +0.00	F/M, NF; +0.00
12-11	L, NF; -1.06	F/M, 18RT; -23.30	F/M, 2RT; +1.53	F/M, 3RT; +2.34
12-12	L, NF; -20.80	F/M, 19RT; -67.59	F/M, 7RT; -10.71	F/M, 8RT; -17.29
12-13	L, NF; -73.44	F/M, 11RT; -24.92	F/M, 9RT; -16.17	F/M, 9RT; -10.90
12-16	L, NF; +31.93	F/M, 13RT; -18.32	F/M, NF; +0.00	F/M, 2RT; +1.07
12-17	L, NF; +46.60	F/M, 12RT; -5.50	F/M, 6RT; +5.57	F/M, 5RT; -1.47
12-18	L, NF; +5.93	F/M, 16RT; -45.21	F/M, 7RT; -7.77	F/M, 7RT; -6.00
12-19	L, NF; -68.70	F/M, 21RT; -48.72	F/M, 2RT; -10.51	F/M, 3RT; -7.08
12-20	L, NF; +18.33	F/M, 17RT; -39.41	F/M, 1RT; -4.96	F/M, 1RT; -2.41
12-23	L, NF; +10.12	F/M, 12RT; -38.83	F/M, 4RT; -2.36	F/M, 5RT; -0.32
12-24	L, NF; +20.19	F/M, 5RT; -36.10	F/M, NF; +0.00	F/M, NF; +0.00
12-26	L, NF; -33.41	F/M, 8RT; -38.63	F/M, 1RT; +0.14	F/M, 1RT; +2.41
12-27	L, NF; +32.86	F/M, 11RT; -53.24	F/M, NF; +0.00	F/M, NF; -2.11
12-30	L, NF; -29.32	F/M, 12RT; -28.81	F/M, 1RT; +0.73	F/M, 1RT; +0.73
12-31	L, NF; +14.58	F/M, 11RT; -10.05	F/M, NF; +0.00	F/M, NF; +0.00
Total	**+25.11**	**-730.40**	**-69.64**	**-75.82**
April 2025				
04-01	L, NF; -10.53	F/M, 13RT; -33.50	F/M, NF; +0.00	F/M, NF; +0.00
04-02	L, NF; +42.65	F/M, 19RT; -63.97	F/M, 1RT; +1.38	F/M, 1RT; -9.69
04-03	L, NF; -35.01	F/M, 31RT; -40.97	F/M, NF; +0.00	F/M, 1RT; -0.94
04-04	L, NF; -126.74	F/M, 36RT; -108.78	F/M, 9RT; -17.31	F/M, 13RT; -40.84
04-07	L, NF; -98.79	F/M, 49RT; -99.90	F/M, 27RT; -74.31	F/M, 27RT; -69.84
04-08	L, NF; +30.71	F/M, 33RT; -67.22	F/M, NF; +0.00	F/M, 2RT; -7.76
04-09	L, NF; +43.84	F/M, 48RT; -196.25	F/M, 16RT; -35.22	F/M, 16RT; -33.64
04-10	L, NF; +78.06	F/M, 36RT; -145.58	F/M, 16RT; -50.82	F/M, 16RT; -61.12
04-11	L, NF; +57.23	F/M, 49RT; -151.21	F/M, 17RT; -16.63	F/M, 16RT; -7.59
04-14	L, NF; +35.17	F/M, 22RT; -44.74	F/M, 11RT; -4.81	F/M, 13RT; -21.99
04-15	L, NF; +54.20	F/M, 19RT; -23.08	F/M, NF; +0.00	F/M, 1RT; +0.32
04-16	L, NF; -75.64	F/M, 26RT; -45.62	F/M, NF; +0.00	F/M, 1RT; -2.60
04-17	L, NF; +22.71	F/M, 19RT; -25.93	F/M, 9RT; -20.24	F/M, 5RT; -14.00
04-18	L, NF; -4.82	F/M, 52RT; -183.09	F/M, 27RT; -85.29	F/M, 27RT; -84.64
04-21	L, NF; -0.61	F/M, 20RT; -31.85	F/M, 1RT; -2.77	F/M, 1RT; +1.90
04-22	L, NF; -32.08	F/M, 25RT; -71.98	F/M, 9RT; -13.07	F/M, 8RT; -6.13
04-23	L, NF; -14.82	F/M, 22RT; -45.89	F/M, 5RT; -15.40	F/M, 5RT; -10.53
04-24	L, NF; +46.08	F/M, 15RT; -6.51	F/M, NF; +0.00	F/M, NF; +0.00
04-25	L, NF; -17.01	F/M, 13RT; -23.75	F/M, 3RT; -16.41	F/M, 2RT; -16.47
04-28	L, NF; +68.55	F/M, 15RT; -48.12	F/M, NF; +0.00	F/M, NF; +0.00
04-29	L, NF; -2.26	F/M, 10RT; -27.93	F/M, 1RT; -1.85	F/M, 1RT; -6.11
04-30	L, NF; -44.94	F/M, 10RT; -23.28	F/M, 1RT; +2.14	F/M, 4RT; -14.26
Total	**+15.95**	**-1509.15**	**-350.61**	**-405.93**

Notes: BH = Buy-and-Hold; BL = Baseline; AR = binary AR regime gate; MS = MS-AR continuous-threshold rule. L = long; F/M = flat/mixed; NF = no flips; RT = approximate round-trip(s). Values are daily PnL in basis points.

8.8 Symbol dictionary

The table below collects the mathematical notation used in the methods sections, in the order in which the symbols are introduced. Where two symbols refer to the same quantity in slightly different parameterisations (for example $c^{(k)}$ versus $\mu^{(k)}$ for the MS-AR(1) intercept), both rows are kept so the dictionary can be used as a lookup reference for either form.

Symbol	Meaning
Data and pre-processing (Section 2).	
$i \in \{A, B\}$	Index of the two legs of a currency pair.
t	Time index of a pre-averaged bar (block of L ticks).
$P_t^{\text{bid}, i}$	Bid price of leg i at time t.
$P_t^{\text{ask}, i}$	Ask price of leg i at time t.
$P_t^{\text{mid}, i}$	Mid-price, $\frac{1}{2}(P_t^{\text{bid}} + P_t^{\text{ask}})$, see (1).
HS_t^i	Basis-point half-spread of leg i at time t, see (1).
L	Pre-averaging block size in number of ticks, see (2).
$\widetilde{P}_k^i$	Pre-averaged mid-price of leg i at block index k, see (2).
r_t^i	Log return of leg i at bar t, $\log P_t^{\text{mid}, i} - \log P_{t-1}^{\text{mid}, i}$.
Baseline pairs-trading rule (Section 3).	
β	Cointegration hedge ratio (global), see (3).
$\beta_t,\ \alpha_t$	Rolling hedge ratio and intercept on window $\mathcal{W}_t$, see (4).
W_β	Length of the rolling OLS window for (β_t, α_t).
$\mathcal{W}_t$	Index set $\{t - W_\beta + 1, \ldots, t\}$ used by the rolling estimator.
$\bar{x}_t,\ \bar{y}_t$	Window means of $x_s = \log P_s^{\text{mid}, B}$ and $y_s = \log P_s^{\text{mid}, A}$.
S_t	Cointegration spread $\log P_t^{\text{mid}, A} - \beta_t \log P_t^{\text{mid}, B} - \alpha_t$.
ΔS_t	Spread return $r_t^A - \beta_{t-1} r_t^B$, see (5).
W_z	Length of the rolling window for the z-score mean and variance.
$\mu_t^S,\ \sigma_t^S$	Rolling mean and standard deviation of S_t over the W_z-bar window.
Z_t	Rolling z-score $(S_t - \mu_t^S)/\sigma_t^S$, see (6).
z_q	Entry threshold on $\lvert Z_t \rvert$ for the baseline rule.
z_x	Exit threshold on $\lvert Z_t \rvert$ for the baseline rule.
$\pi_t \in \{-1, 0, +1\}$	Position at bar t (short / flat / long the spread), see (7).
π_t^{BH}	Buy-and-Hold reference position, $\pi_t^{\text{BH}} \equiv +1$.
π_t^{baseline}	Baseline-rule position produced by (7).
Regime-aware models (Section 4).	
p	Order of the generic AR(p) model in (8).
$c,\ \phi_j,\ \sigma^2$	Intercept, AR coefficients, and innovation variance of AR(p); for $p = 1$, $\phi_1 \equiv \phi$, see (9).
ε_t	Innovation in the AR(p)/AR(1) model, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$.
K	Number of regimes in the Markov chain.

Symbol	Meaning
$K_t \in \{1,\dots,K\}$	Hidden regime at bar t.
$\mathbf{P} = (p_{ij})$	$K \times K$ transition matrix, $p_{ij} = \mathbb{P}(K_t = j \mid K_{t-1} = i)$, see (10).
$\nu = (\nu_1,\dots,\nu_K)$	Initial-state distribution of the Markov chain.
$c^{(k)}$, $\rho^{(k)}$, $\sigma^{(k)}$	Regime-k intercept, AR(1) coefficient, and innovation standard deviation in (11).
$\mu^{(k)}$	Regime-k unconditional mean, $\mu^{(k)} = c^{(k)}/(1-\rho^{(k)})$, see (12).
Θ	Full HMM parameter set $\{\nu, \mathbf{P}, c^{(k)}, \rho^{(k)}, \sigma^{(k)}\}_{k=1}^{K}$, see (13).
$\gamma_t(k)$	Smoothed regime posterior $\mathbb{P}(K_t = k \mid S_1,\dots,S_T;\Theta)$, see (14).
k_{MR}, k_{DR}	Indices of the mean-reverting and drifting regimes, by innovation variance, see (15).
γ_t^{MR}, γ_t^{DR}	Posterior probabilities of the MR and DR regimes at bar t; $\gamma_t^{\mathrm{MR}} + \gamma_t^{\mathrm{DR}} = 1$.
$\delta \in (0,1)$	Danger-threshold tolerance in the AR gate, see (16).
$G_t \in \{0,1\}$	Binary AR gate, $G_t = \mathbb{1}\{\gamma_t^{\mathrm{MR}} \geq 1-\delta\}$.
z_v	MS-AR drifting-regime threshold, with $z_v \geq z_q$, see (18).
$z_q^{\mathrm{eff}}(t)$	Regime-weighted effective entry threshold for MS-AR, see (18).
Back-test protocol and performance metrics (Section 5).	
R_t	Strategy return at bar t.
μ_R, σ_R	Mean and standard deviation of R_t.
R_f	Benchmark risk-free rate, set to zero throughout.
T	Number of bars per year used in the annualisation of σ_R.
σ_{ann}	Annualised return volatility, $\sigma_R\sqrt{T}$, see (19).
$\mathcal{S}$, $\mathcal{K}$	Skewness and kurtosis of R_t, see (19).
α	Confidence level for VaR/CVaR, $\alpha = 0.95$, see (20).
VaR_α	Value-at-Risk at level α, $-Q_{1-\alpha}(R)$.
CVaR_α	Conditional VaR (expected shortfall) at level α.
TF	Tail-fatness ratio $\mathrm{CVaR}_\alpha/\mathrm{VaR}_\alpha$, see (20).
V_t, $V_t^\star$	Portfolio value at bar t and its running peak $\max_{s\leq t} V_s$.
D_t	Drawdown $(V_t^\star - V_t)/V_t^\star$, see (21).
MDD, UI	Maximum drawdown and Ulcer index, see (21).
SR	Sharpe ratio $(\mu_R - R_f)/\sigma_R$, see (22).
Sortino	Sortino ratio $(\mu_R - R_f)/\sigma_d$, with σ_d the downside deviation.
Calmar	Calmar ratio, annualised return divided by maximum drawdown.
AR	Annualised return used in the Calmar ratio. (Not to be confused with the AR-gated strategy.)